

STAR Grafana Dashboard JSON

Source Listing

Auto-generated syntax-highlighted listing

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STAR Development Team
Locked Inc.

Generated by `scripts/codedump-pdf.sh`.

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1 (root)

grafana-star-dashboard.json

```
1 {
2   "apiVersion": "dashboard.grafana.app/v2",
3   "kind": "Dashboard",
4   "metadata": {
5     "name": "star-robot-telemetry",
6     "namespace": "default",
7     "uid": "3263aaa7-51c4-4d1a-8786-c86f962177bc",
8     "resourceVersion": "1776950319890995",
9     "generation": 25,
10    "creationTimestamp": "2026-04-23T07:12:11Z",
11    "labels": {
12      "grafana.app/deprecatedInternalID": "2577689642606592"
13    },
14    "annotations": {
15      "grafana.app/createdBy": "user:dfi7zlylxemtce",
16      "grafana.app/message": "STOP%: debounced live-push on input change (no need to click Apply)",
17      "grafana.app/updatedBy": "user:dfi7zlylxemtce",
18      "grafana.app/updatedTimestamp": "2026-04-23T13:18:39Z",
19      "grafana.app/folder": ""
20    }
21  },
22  "spec": {
23    "annotations": [
24      {
25        "kind": "AnnotationQuery",
26        "spec": {
27          "query": {
28            "kind": "DataQuery",
29            "group": "grafana",
30            "version": "v0",
31            "datasource": {
32              "name": "-- Grafana --"
33            },
34            "spec": {}
35          },
36          "enable": true,
37          "hide": true,
38          "iconColor": "rgba(0, 211, 255, 1)",
39          "name": "Annotations & Alerts",
40          "builtIn": true,
41          "legacyOptions": {
42            "type": "dashboard"
43          }
44        }
45      ]
46    },
47    "cursorSync": "Crosshair",
48    "editable": true,
49    "elements": {
50      "panel-1": {
51        "kind": "Panel",
52        "spec": {
53          "id": 1,
54          "title": "FL velocity",
55          "description": "",
56          "links": [],
57          "data": {
58            "kind": "QueryGroup",
59            "spec": {
60              "queries": [
61                {
62                  "kind": "PanelQuery",
63                  "spec": {
64                    "query": {
65                      "kind": "DataQuery",
66                      "group": "",
67                      "version": "v0",
68                      "spec": {
69                        "expr": "star_motor_velocity_mps{wheel=\"fl\"}",
70                        "legendFormat": ""
71                      }
72                    },
73                    "refId": "A",
74                    "hidden": false
75                  }
76                }
77              ]
78            }
79          }
80        }
81      }
82    }
83  }
84 }
```

```

76     }
77   ],
78   "transformations": [],
79   "queryOptions": {}
80 }
81 },
82 "vizConfig": {
83   "kind": "VizConfig",
84   "group": "gauge",
85   "version": "",
86   "spec": {
87     "options": {
88       "orientation": "auto",
89       "showThresholdLabels": false,
90       "showThresholdMarkers": true
91     },
92     "fieldConfig": {
93       "defaults": {
94         "unit": "velocityms",
95         "min": -0.3,
96         "max": 0.3,
97         "thresholds": {
98           "mode": "absolute",
99           "steps": [
100             {
101               "value": null,
102               "color": "blue"
103             },
104             {
105               "value": -0.25,
106               "color": "green"
107             },
108             {
109               "value": 0.2,
110               "color": "yellow"
111             },
112             {
113               "value": 0.25,
114               "color": "red"
115             }
116           ]
117         }
118       },
119       "overrides": []
120     }
121   }
122 },
123 },
124 },
125 "panel-10": {
126   "kind": "Panel",
127   "spec": {
128     "id": 10,
129     "title": "Wheel duty over time",
130     "description": "",
131     "links": [],
132     "data": {
133       "kind": "QueryGroup",
134       "spec": {
135         "queries": [
136           {
137             "kind": "PanelQuery",
138             "spec": {
139               "query": {
140                 "kind": "DataQuery",
141                 "group": "",
142                 "version": "v0",
143                 "spec": {
144                   "expr": "star_motor_duty_percent",
145                   "legendFormat": "{{wheel}}"
146                 }
147               },
148               "refId": "A",
149               "hidden": false
150             }
151           }
152         ],
153         "transformations": [],
154         "queryOptions": {}

```

```

155     }
156   },
157   "vizConfig": {
158     "kind": "VizConfig",
159     "group": "timeseries",
160     "version": "",
161     "spec": {
162       "options": {},
163       "fieldConfig": {
164         "defaults": {
165           "unit": "percent"
166         },
167         "overrides": []
168       }
169     }
170   }
171 },
172 },
173 "panel-11": {
174   "kind": "Panel",
175   "spec": {
176     "id": 11,
177     "title": "Motor faults (any != 0 is bad)",
178     "description": "",
179     "links": [],
180     "data": {
181       "kind": "QueryGroup",
182       "spec": {
183         "queries": [
184           {
185             "kind": "PanelQuery",
186             "spec": {
187               "query": {
188                 "kind": "DataQuery",
189                 "group": "",
190                 "version": "v0",
191                 "spec": {
192                   "expr": "star_motor_fault_byte",
193                   "legendFormat": "{{wheel}}"
194                 }
195               },
196               "refId": "A",
197               "hidden": false
198             }
199           }
200         ],
201         "transformations": [],
202         "queryOptions": {}
203       }
204     },
205     "vizConfig": {
206       "kind": "VizConfig",
207       "group": "stat",
208       "version": "",
209       "spec": {
210         "options": {
211           "colorMode": "background",
212           "graphMode": "none",
213           "textMode": "value_and_name"
214         },
215         "fieldConfig": {
216           "defaults": {
217             "thresholds": {
218               "mode": "absolute",
219               "steps": [
220                 {
221                   "value": null,
222                   "color": "green"
223                 },
224                 {
225                   "value": 1,
226                   "color": "red"
227                 }
228               ]
229             }
230           },
231           "overrides": []
232         }
233       }
234     }
235   }
236 }

```

```

234     }
235   }
236 },
237 "panel-12": {
238   "kind": "Panel",
239   "spec": {
240     "id": 12,
241     "title": "Roll",
242     "description": "",
243     "links": [],
244     "data": {
245       "kind": "QueryGroup",
246       "spec": {
247         "queries": [
248           {
249             "kind": "PanelQuery",
250             "spec": {
251               "query": {
252                 "kind": "DataQuery",
253                 "group": "",
254                 "version": "v0",
255                 "spec": {
256                   "expr": "star_imu_orientation_deg{axis=\"roll\"}"
257                 }
258               },
259               "refId": "A",
260               "hidden": false
261             }
262           }
263         ],
264         "transformations": [],
265         "queryOptions": {}
266       }
267     },
268     "vizConfig": {
269       "kind": "VizConfig",
270       "group": "gauge",
271       "version": "",
272       "spec": {
273         "options": {},
274         "fieldConfig": {
275           "defaults": {
276             "unit": "degree",
277             "min": -180,
278             "max": 180
279           },
280           "overrides": []
281         }
282       }
283     }
284   },
285   "panel-13": {
286     "kind": "Panel",
287     "spec": {
288       "id": 13,
289       "title": "Pitch",
290       "description": "",
291       "links": [],
292       "data": {
293         "kind": "QueryGroup",
294         "spec": {
295           "queries": [
296             {
297               "kind": "PanelQuery",
298               "spec": {
299                 "query": {
300                   "kind": "DataQuery",
301                   "group": "",
302                   "version": "v0",
303                   "spec": {
304                     "expr": "star_imu_orientation_deg{axis=\"pitch\"}"
305                   }
306                 },
307                 "refId": "A",
308                 "hidden": false
309               }
310             }
311           ]
312         }

```

```

313         "transformations": [],
314         "queryOptions": {}
315     }
316 },
317 "vizConfig": {
318     "kind": "VizConfig",
319     "group": "gauge",
320     "version": "",
321     "spec": {
322         "options": {},
323         "fieldConfig": {
324             "defaults": {
325                 "unit": "degree",
326                 "min": -180,
327                 "max": 180
328             },
329             "overrides": []
330         }
331     }
332 }
333 },
334 },
335 "panel-14": {
336     "kind": "Panel",
337     "spec": {
338         "id": 14,
339         "title": "Heading",
340         "description": "",
341         "links": [],
342         "data": {
343             "kind": "QueryGroup",
344             "spec": {
345                 "queries": [
346                     {
347                         "kind": "PanelQuery",
348                         "spec": {
349                             "query": {
350                                 "kind": "DataQuery",
351                                 "group": "",
352                                 "version": "v0",
353                                 "spec": {
354                                     "expr": "star_imu_orientation_deg{axis=\"heading\"}"
355                                 }
356                             },
357                             "refId": "A",
358                             "hidden": false
359                         }
360                     }
361                 ],
362                 "transformations": [],
363                 "queryOptions": {}
364             }
365         },
366         "vizConfig": {
367             "kind": "VizConfig",
368             "group": "gauge",
369             "version": "",
370             "spec": {
371                 "options": {},
372                 "fieldConfig": {
373                     "defaults": {
374                         "unit": "degree",
375                         "min": 0,
376                         "max": 360
377                     },
378                     "overrides": []
379                 }
380             }
381         }
382     }
383 },
384 "panel-15": {
385     "kind": "Panel",
386     "spec": {
387         "id": 15,
388         "title": "Gyro (deg/s)",
389         "description": "",
390         "links": [],
391         "data": {

```

```

392     "kind": "QueryGroup",
393     "spec": {
394       "queries": [
395         {
396           "kind": "PanelQuery",
397           "spec": {
398             "query": {
399               "kind": "DataQuery",
400               "group": "",
401               "version": "v0",
402               "spec": {
403                 "expr": "star_imu_gyro_dps",
404                 "legendFormat": "{{axis}}"
405               }
406             },
407             "refId": "A",
408             "hidden": false
409           }
410         }
411       ],
412       "transformations": [],
413       "queryOptions": {}
414     }
415   },
416   "vizConfig": {
417     "kind": "VizConfig",
418     "group": "timeseries",
419     "version": "",
420     "spec": {
421       "options": {},
422       "fieldConfig": {
423         "defaults": {
424           "unit": "rotdegs"
425         },
426         "overrides": []
427       }
428     }
429   }
430 },
431 },
432 "panel-16": {
433   "kind": "Panel",
434   "spec": {
435     "id": 16,
436     "title": "Linear acceleration (m/s^2)",
437     "description": "",
438     "links": [],
439     "data": {
440       "kind": "QueryGroup",
441       "spec": {
442         "queries": [
443           {
444             "kind": "PanelQuery",
445             "spec": {
446               "query": {
447                 "kind": "DataQuery",
448                 "group": "",
449                 "version": "v0",
450                 "spec": {
451                   "expr": "star_imu_accel_mps2",
452                   "legendFormat": "{{axis}}"
453                 }
454               },
455               "refId": "A",
456               "hidden": false
457             }
458           }
459         ],
460         "transformations": [],
461         "queryOptions": {}
462       }
463     },
464     "vizConfig": {
465       "kind": "VizConfig",
466       "group": "timeseries",
467       "version": "",
468       "spec": {
469         "options": {},
470         "fieldConfig": {

```



```

471         "defaults": {
472             "unit": "accMS2"
473         },
474         "overrides": []
475     }
476 }
477 }
478 }
479 },
480 "panel-17": {
481     "kind": "Panel",
482     "spec": {
483         "id": 17,
484         "title": "Linear velocity",
485         "description": "",
486         "links": [],
487         "data": {
488             "kind": "QueryGroup",
489             "spec": {
490                 "queries": [
491                     {
492                         "kind": "PanelQuery",
493                         "spec": {
494                             "query": {
495                                 "kind": "DataQuery",
496                                 "group": "",
497                                 "version": "v0",
498                                 "spec": {
499                                     "expr": "star_odom_linear_mps"
500                                 }
499                             },
501                             "refId": "A",
502                             "hidden": false
503                         }
504                     }
505                 ],
506                 "transformations": [],
507                 "queryOptions": {}
508             }
509         },
510     },
511     "vizConfig": {
512         "kind": "VizConfig",
513         "group": "gauge",
514         "version": "",
515         "spec": {
516             "options": {},
517             "fieldConfig": {
518                 "defaults": {
519                     "unit": "velocityms",
520                     "min": -0.3,
521                     "max": 0.3
522                 },
523                 "overrides": []
524             }
525         }
526     }
527 },
528 },
529 "panel-18": {
530     "kind": "Panel",
531     "spec": {
532         "id": 18,
533         "title": "Angular velocity",
534         "description": "",
535         "links": [],
536         "data": {
537             "kind": "QueryGroup",
538             "spec": {
539                 "queries": [
540                     {
541                         "kind": "PanelQuery",
542                         "spec": {
543                             "query": {
544                                 "kind": "DataQuery",
545                                 "group": "",
546                                 "version": "v0",
547                                 "spec": {
548                                     "expr": "star_odom_angular_radps"
549                                 }

```

```

550         },
551         "refId": "A",
552         "hidden": false
553     }
554 }
555 ],
556 "transformations": [],
557 "queryOptions": {}
558 }
559 },
560 "vizConfig": {
561     "kind": "VizConfig",
562     "group": "gauge",
563     "version": "",
564     "spec": {
565         "options": {},
566         "fieldConfig": {
567             "defaults": {
568                 "unit": "rads",
569                 "min": -2,
570                 "max": 2
571             },
572             "overrides": []
573         }
574     }
575 }
576 },
577 "panel-19": {
578     "kind": "Panel",
579     "spec": {
580         "id": 19,
581         "title": "Odom pose (x, y, theta)",
582         "description": "",
583         "links": [],
584         "data": {
585             "kind": "QueryGroup",
586             "spec": {
587                 "queries": [
588                     {
589                         "kind": "PanelQuery",
590                         "spec": {
591                             "query": {
592                                 "kind": "DataQuery",
593                                 "group": "",
594                                 "version": "v0",
595                                 "spec": {
596                                     "expr": "star_odom_pose_x_m",
597                                     "legendFormat": "x (m)"
598                                 }
599                             },
600                             "refId": "A",
601                             "hidden": false
602                         }
603                     },
604                     {
605                         "kind": "PanelQuery",
606                         "spec": {
607                             "query": {
608                                 "kind": "DataQuery",
609                                 "group": "",
610                                 "version": "v0",
611                                 "spec": {
612                                     "expr": "star_odom_pose_y_m",
613                                     "legendFormat": "y (m)"
614                                 }
615                             },
616                             "refId": "B",
617                             "hidden": false
618                         }
619                     },
620                     {
621                         "kind": "PanelQuery",
622                         "spec": {
623                             "query": {
624                                 "kind": "DataQuery",
625                                 "group": "",
626                                 "version": "v0",
627                                 "spec": {

```

```

629         "expr": "star_odom_pose_theta_rad",
630         "legendFormat": "theta (rad)"
631     }
632 },
633     "refId": "C",
634     "hidden": false
635 }
636 },
637 ],
638 "transformations": [],
639 "queryOptions": {}
640 }
641 },
642 "vizConfig": {
643     "kind": "VizConfig",
644     "group": "stat",
645     "version": "",
646     "spec": {
647         "options": {
648             "colorMode": "value",
649             "graphMode": "area",
650             "textMode": "value_and_name"
651         },
652         "fieldConfig": {
653             "defaults": {},
654             "overrides": []
655         }
656     }
657 }
658 },
659 },
660 "panel-2": {
661     "kind": "Panel",
662     "spec": {
663         "id": 2,
664         "title": "FR velocity",
665         "description": "",
666         "links": [],
667         "data": {
668             "kind": "QueryGroup",
669             "spec": {
670                 "queries": [
671                     {
672                         "kind": "PanelQuery",
673                         "spec": {
674                             "query": {
675                                 "kind": "DataQuery",
676                                 "group": "",
677                                 "version": "v0",
678                                 "spec": {
679                                     "expr": "star_motor_velocity_mps{wheel=\"fr\"}",
680                                     "legendFormat": ""
681                                 }
682                             },
683                             "refId": "A",
684                             "hidden": false
685                         }
686                     }
687                 ],
688                 "transformations": [],
689                 "queryOptions": {}
690             }
691         },
692         "vizConfig": {
693             "kind": "VizConfig",
694             "group": "gauge",
695             "version": "",
696             "spec": {
697                 "options": {},
698                 "fieldConfig": {
699                     "defaults": {
700                         "unit": "velocityms",
701                         "min": -0.3,
702                         "max": 0.3,
703                         "thresholds": {
704                             "mode": "absolute",
705                             "steps": [
706                                 {
707                                     "value": null,

```

```

708         "color": "blue"
709     },
710     {
711         "value": -0.25,
712         "color": "green"
713     },
714     {
715         "value": 0.2,
716         "color": "yellow"
717     },
718     {
719         "value": 0.25,
720         "color": "red"
721     }
722 ]
723 }
724 },
725 "overrides": []
726 }
727 }
728 }
729 }
730 },
731 "panel-20": {
732     "kind": "Panel",
733     "spec": {
734         "id": 20,
735         "title": "Line rate by kind (msg/s)",
736         "description": "",
737         "links": [],
738         "data": {
739             "kind": "QueryGroup",
740             "spec": {
741                 "queries": [
742                     {
743                         "kind": "PanelQuery",
744                         "spec": {
745                             "query": {
746                                 "kind": "DataQuery",
747                                 "group": "",
748                                 "version": "v0",
749                                 "spec": {
750                                     "expr": "rate(star_bridge_rx_lines_total[1m])",
751                                     "legendFormat": "{{kind}}"
752                                 }
753                             },
754                             "refId": "A",
755                             "hidden": false
756                         }
757                     }
758                 ],
759                 "transformations": [],
760                 "queryOptions": {}
761             }
762         },
763         "vizConfig": {
764             "kind": "VizConfig",
765             "group": "timeseries",
766             "version": "",
767             "spec": {
768                 "options": {},
769                 "fieldConfig": {
770                     "defaults": {},
771                     "overrides": []
772                 }
773             }
774         }
775     },
776 },
777 "panel-21": {
778     "kind": "Panel",
779     "spec": {
780         "id": 21,
781         "title": "Parse errors (/s)",
782         "description": "",
783         "links": [],
784         "data": {
785             "kind": "QueryGroup",
786             "spec": {

```

```

787     "queries": [
788     {
789         "kind": "PanelQuery",
790         "spec": {
791             "query": {
792                 "kind": "DataQuery",
793                 "group": "",
794                 "version": "v0",
795                 "spec": {
796                     "expr": "rate(star_bridge_parse_errors_total[1m])",
797                     "legendFormat": "{{kind}}"
798                 }
799             },
800             "refId": "A",
801             "hidden": false
802         }
803     },
804     ],
805     "transformations": [],
806     "queryOptions": {}
807 },
808 },
809 "vizConfig": {
810     "kind": "VizConfig",
811     "group": "timeseries",
812     "version": "",
813     "spec": {
814         "options": {},
815         "fieldConfig": {
816             "defaults": {
817                 "color": {
818                     "mode": "fixed",
819                     "fixedColor": "red"
820                 }
821             },
822             "overrides": []
823         }
824     }
825 },
826 },
827 },
828 "panel-22": {
829     "kind": "Panel",
830     "spec": {
831         "id": 22,
832         "title": "Serial reopens (cumulative)",
833         "description": "",
834         "links": [],
835         "data": {
836             "kind": "QueryGroup",
837             "spec": {
838                 "queries": [
839                     {
840                         "kind": "PanelQuery",
841                         "spec": {
842                             "query": {
843                                 "kind": "DataQuery",
844                                 "group": "",
845                                 "version": "v0",
846                                 "spec": {
847                                     "expr": "star_bridge_serial_reopens_total"
848                                 }
849                             },
850                             "refId": "A",
851                             "hidden": false
852                         }
853                     }
854                 ],
855                 "transformations": [],
856                 "queryOptions": {}
857             }
858         },
859         "vizConfig": {
860             "kind": "VizConfig",
861             "group": "stat",
862             "version": "",
863             "spec": {
864                 "options": {},
865                 "fieldConfig": {

```

```

866         "defaults": {},
867         "overrides": []
868     }
869 }
870 }
871 }
872 },
873 "panel-23": {
874     "kind": "Panel",
875     "spec": {
876         "id": 23,
877         "title": "RX bytes total",
878         "description": "",
879         "links": [],
880         "data": {
881             "kind": "QueryGroup",
882             "spec": {
883                 "queries": [
884                     {
885                         "kind": "PanelQuery",
886                         "spec": {
887                             "query": {
888                                 "kind": "DataQuery",
889                                 "group": "",
890                                 "version": "v0",
891                                 "spec": {
892                                     "expr": "star_bridge_rx_bytes_total"
893                                 }
894                             },
895                             "refId": "A",
896                             "hidden": false
897                         }
898                     ]
899                 },
900                 "transformations": [],
901                 "queryOptions": {}
902             }
903         },
904         "vizConfig": {
905             "kind": "VizConfig",
906             "group": "stat",
907             "version": "",
908             "spec": {
909                 "options": {},
910                 "fieldConfig": {
911                     "defaults": {
912                         "unit": "bytes"
913                     },
914                     "overrides": []
915                 }
916             }
917         }
918     },
919 },
920 "panel-24": {
921     "kind": "Panel",
922     "spec": {
923         "id": 24,
924         "title": "Closest obstacle",
925         "description": "",
926         "links": [],
927         "data": {
928             "kind": "QueryGroup",
929             "spec": {
930                 "queries": [
931                     {
932                         "kind": "PanelQuery",
933                         "spec": {
934                             "query": {
935                                 "kind": "DataQuery",
936                                 "group": "",
937                                 "version": "v0",
938                                 "spec": {
939                                     "expr": "star_scan_closest_range_m"
940                                 }
941                             },
942                             "refId": "A",
943                             "hidden": false
944                         }

```

```

945     }
946   ],
947   "transformations": [],
948   "queryOptions": {}
949 }
950 },
951 "vizConfig": {
952   "kind": "VizConfig",
953   "group": "gauge",
954   "version": "",
955   "spec": {
956     "options": {},
957     "fieldConfig": {
958       "defaults": {
959         "unit": "lengthm",
960         "min": 0,
961         "max": 3,
962         "thresholds": {
963           "mode": "absolute",
964           "steps": [
965             {
966               "value": null,
967               "color": "red"
968             },
969             {
970               "value": 0.5,
971               "color": "yellow"
972             },
973             {
974               "value": 1,
975               "color": "green"
976             }
977           ]
978         }
979       },
980       "overrides": []
981     }
982   }
983 }
984 },
985 },
986 "panel-25": {
987   "kind": "Panel",
988   "spec": {
989     "id": 25,
990     "title": "Closest bearing (0deg=forward)",
991     "description": "",
992     "links": [],
993     "data": {
994       "kind": "QueryGroup",
995       "spec": {
996         "queries": [
997           {
998             "kind": "PanelQuery",
999             "spec": {
1000               "query": {
1001                 "kind": "DataQuery",
1002                 "group": "",
1003                 "version": "v0",
1004                 "spec": {
1005                   "expr": "star_scan_closest_bearing_deg"
1006                 }
1007               },
1008               "refId": "A",
1009               "hidden": false
1010             }
1011           }
1012         ],
1013         "transformations": [],
1014         "queryOptions": {}
1015       }
1016     },
1017     "vizConfig": {
1018       "kind": "VizConfig",
1019       "group": "gauge",
1020       "version": "",
1021       "spec": {
1022         "options": {},
1023         "fieldConfig": {

```

```

1024         "defaults": {
1025             "unit": "degree",
1026             "min": -180,
1027             "max": 180
1028         },
1029         "overrides": []
1030     }
1031 }
1032 }
1033 }
1034 },
1035 "panel-26": {
1036     "kind": "Panel",
1037     "spec": {
1038         "id": 26,
1039         "title": "Obstacles < 1 m",
1040         "description": "",
1041         "links": [],
1042         "data": {
1043             "kind": "QueryGroup",
1044             "spec": {
1045                 "queries": [
1046                     {
1047                         "kind": "PanelQuery",
1048                         "spec": {
1049                             "query": {
1050                                 "kind": "DataQuery",
1051                                 "group": "",
1052                                 "version": "v0",
1053                                 "spec": {
1054                                     "expr": "star_scan_obstacles_under_1m"
1055                                 }
1056                             },
1057                             "refId": "A",
1058                             "hidden": false
1059                         }
1060                     }
1061                 ],
1062                 "transformations": [],
1063                 "queryOptions": {}
1064             }
1065         },
1066         "vizConfig": {
1067             "kind": "VizConfig",
1068             "group": "stat",
1069             "version": "",
1070             "spec": {
1071                 "options": {
1072                     "colorMode": "background",
1073                     "graphMode": "area"
1074                 },
1075                 "fieldConfig": {
1076                     "defaults": {
1077                         "thresholds": {
1078                             "mode": "absolute",
1079                             "steps": [
1080                                 {
1081                                     "value": null,
1082                                     "color": "green"
1083                                 },
1084                                 {
1085                                     "value": 1,
1086                                     "color": "yellow"
1087                                 },
1088                                 {
1089                                     "value": 10,
1090                                     "color": "red"
1091                                 }
1092                             ]
1093                         }
1094                     },
1095                     "overrides": []
1096                 }
1097             }
1098         }
1099     },
1100 },
1101 "panel-27": {
1102     "kind": "Panel",

```



```

1103     "spec": {
1104         "id": 27,
1105         "title": "Scan range min / mean / max",
1106         "description": "",
1107         "links": [],
1108         "data": {
1109             "kind": "QueryGroup",
1110             "spec": {
1111                 "queries": [
1112                     {
1113                         "kind": "PanelQuery",
1114                         "spec": {
1115                             "query": {
1116                                 "kind": "DataQuery",
1117                                 "group": "",
1118                                 "version": "v0",
1119                                 "spec": {
1120                                     "expr": "star_scan_range_m",
1121                                     "legendFormat": "{{stat}}"
1122                                 }
1123                             },
1124                             "refId": "A",
1125                             "hidden": false
1126                         }
1127                     }
1128                 ],
1129                 "transformations": [],
1130                 "queryOptions": {}
1131             }
1132         },
1133         "vizConfig": {
1134             "kind": "VizConfig",
1135             "group": "stat",
1136             "version": "",
1137             "spec": {
1138                 "options": {
1139                     "colorMode": "value",
1140                     "textMode": "value_and_name"
1141                 },
1142                 "fieldConfig": {
1143                     "defaults": {
1144                         "unit": "lengthm"
1145                     },
1146                     "overrides": []
1147                 }
1148             }
1149         }
1150     },
1151     "panel-29": {
1152         "kind": "Panel",
1153         "spec": {
1154             "id": 29,
1155             "title": "Cockpit -- map + controls + teleop",
1156             "description": "",
1157             "links": [],
1158             "data": {
1159                 "kind": "QueryGroup",
1160                 "spec": {
1161                     "queries": [],
1162                     "transformations": [],
1163                     "queryOptions": {}
1164                 }
1165             },
1166             "vizConfig": {
1167                 "kind": "VizConfig",
1168                 "group": "text",
1169                 "version": "",
1170                 "spec": {
1171                     "options": {

```

```

"content":
< "style">\n.star-cockpit{display:flex;flex-direction:column;height:100%;gap:8px;background:#111;padding:8px;box-sizing:border
b{font:600 12px system-ui;color:#888;margin-right:4px;min-width:72px}\n.star-btn{padding:6px
12px;border-radius:4px;border:1px solid #444;background:#222;color:#ccc;text-decoration:none;font:600 13px
system-ui;cursor:pointer;white-space:nowrap;user-select:none}\n.star-btn:hover{background:#333}\n.star-btn.danger{background
10px;border-radius:20px;border:1px solid
#444;background:#222;cursor:pointer;user-select:none;min-width:130px;justify-content:space-between}\n.star-switch:hover{bac
.knob{width:38px;height:20px;border-radius:12px;background:#444;position:relative;transition:background
.15s}\n.star-switch
.knob::after{content:"";position:absolute;top:2px;left:2px;width:16px;height:16px;border-radius:50%;background:#ddd;trans
.15s}\n.star-switch.on .knob{background:#0a6}\n.star-switch.on
.knob::after{left:20px}\n.star-switch.estop.on .knob{background:#c22}\n.star-switch .label{font:600 13px
system-ui}\n.star-switch.on.estop .label{color:#f88}\n.star-switch.on:not(.estop)
.label{color:#8f8}\n.star-num{display:inline-flex;align-items:center;gap:6px}\n.star-num
input[type=number]{width:80px;padding:4px 6px;border-radius:4px;border:1px solid
#444;background:#222;color:#eee;font:13px system-ui}\n.star-num span{font:600 11px
system-ui;color:#888}\n.star-main{flex:1;display:flex;gap:8px;min-height:0}\n.star-map-img{flex:1;display:flex;justify-cont
img{max-width:100%;max-height:100%;image-rendering:pixelated}\n.star-map-img
img[src=""]{display:none}\n.star-teleop{display:grid;grid-template-columns:60px 60px
60px;grid-template-rows:60px 60px 60px;gap:4px;align-self:center}\n.star-tele-btn{background:#222;border:1px
solid #444;border-radius:6px;color:#ccc;font:700 20px
system-ui;cursor:pointer;user-select:none;display:flex;align-items:center;justify-content:center}\n.star-tele-btn:hover{bac
(pointer: coarse), (max-width: 820px){ .star-tele-btn.corner{display:none} }}\n#star-status{font:12px
monospace;color:#888;margin-left:auto}\n.star-hint{font:11px
system-ui;color:#666;margin-left:8px}\n</style>\n<div class=\n.star-cockpit\n tabindex=\n0\n
id=\nstar-cockpit\n">\n <div class=\nstar-row\n">\n <b>CONTROL</b>\n <label class=\nstar-switch estop\n
id=\nstar-sw-estop\n"><span class=\nlabel\n">E-STOP</span><span class=\nknob\n"></span></label>\n <label
class=\nstar-switch\n id=\nstar-sw-auto\n"><span class=\nlabel\n">MANUAL</span><span
class=\nknob\n"></span></label>\n <span class=\nstar-num\n">\n <span>SPEED</span>\n <input
type=\nnumber\n id=\nstar-speed\n min=\n0\n max=\n100\n step=\n5\n value=\n50\n">\n <span>%</span>\n
<button class=\nstar-btn\n id=\nstar-speed-apply\n">Apply</button>\n </span>\n <span
class=\nstar-num\n">\n <span>TURN</span>\n <input type=\nnumber\n id=\nstar-turn\n min=\n0.5\n
max=\n8\n step=\n0.5\n value=\n4\n">\n <span>rad/s</span>\n <button class=\nstar-btn\n
id=\nstar-turn-apply\n">Apply</button>\n </span>\n <span class=\nstar-hint\n">click panel then WASD /
arrows to drive</span>\n <span id=\nstar-status\n">loading...</span>\n </div>\n <div
class=\nstar-row\n">\n <b>MAP</b>\n <a class=\nstar-btn\n
href=\nhttps://bsikar:y0hj8pQBG1zbpYrDTAG9UK@robot.flavorblast.sbs/map.pgm\n download=\nmap.pgm\n">Download
.pgm</a>\n <a class=\nstar-btn\n
href=\nhttps://bsikar:y0hj8pQBG1zbpYrDTAG9UK@robot.flavorblast.sbs/map.yaml\n download=\nmap.yaml\n">Download
.yaml</a>\n <button class=\nstar-btn\n id=\nstar-map-record\n title=\nStart/stop SLAM
mapping\n">loading...</button>\n <button class=\nstar-btn danger\n id=\nstar-map-clear\n title=\nWipe the
map (preserves current mapping state)\n">CLEAR MAP</button>\n <button class=\nstar-btn\n
id=\nstar-explore\n title=\nAuto-explore the area until no frontiers remain (requires mapping
active)\n">loading...</button>\n <span class=\nstar-num\n title=\nHow patient to be before declaring
exploration done (higher = wait longer for growth)\n">\n <span>STOP%</span>\n <input
type=\nnumber\n id=\nstar-explore-th\n min=\n0\n max=\n100\n step=\n5\n value=\n50\n">\n
<span>%</span>\n <button class=\nstar-btn\n id=\nstar-explore-th-apply\n">Apply</button>\n </span>\n
</div>\n <div class=\nstar-row\n style=\nfont:11px system-ui;color:#9a9;padding:2px 0;border-left:2px solid
#363;padding-left:10px;gap:18px\n">\n <b style=\ncolor:#666\n">HINTS</b>\n <span><b
style=\ncolor:#bc8\n">MANUAL (WASD/arrows):</b> SPEED 50% first / 100% comfy * TURN 3-4 rad/s</span>\n
<span><b style=\ncolor:#8bc\n">AUTO goals (click in 3D):</b> SPEED 60% * TURN 1-2 rad/s</span>\n <span><b
style=\ncolor:#c8b\n">EXPLORE (auto-map):</b> start MAPPING first, then EXPLORE. STOP% = patience (higher
waits longer before calling it done)</span>\n </div>\n<div class=\nstar-main\n">\n <div
class=\nstar-map-img\n"><img id=\nstar-map-img\n src=\n alt=\n"></div>\n <div class=\nstar-teleop\n
id=\nstar-teleop\n">\n <div class=\nstar-tele-btn ne corner\n data-dir=\nup-left\n">NW</div>\n <div
class=\nstar-tele-btn up\n data-dir=\nup\n">N</div>\n <div class=\nstar-tele-btn nw corner\n
data-dir=\nup-right\n">NE</div>\n <div class=\nstar-tele-btn left\n data-dir=\nleft\n">W</div>\n
<div class=\nstar-tele-btn right\n data-dir=\nright\n">E</div>\n <div class=\nstar-tele-btn sw corner\n
data-dir=\ndown-left\n">SW</div>\n <div class=\nstar-tele-btn down\n data-dir=\ndown\n">S</div>\n
<div class=\nstar-tele-btn se corner\n data-dir=\ndown-right\n">SE</div>\n </div>\n
</div>\n</div>\n<script>\n(function(){\n const AUTH = \nBasic YnNpa2FyOnkwaGo4cFFCRzF6YnB5c2RkUQUc5VUs=\n;\n
const _fetch = window.fetch.bind(window);\n function fetch(url, opts){ opts = opts||{}; opts.headers =
Object.assign({}, opts.headers||{}, {Authorization: AUTH}); return _fetch(url, opts); }\n const API =
\nhttps://robot.flavorblast.sbs/api\n;\n const PUBLIC = \nhttps://robot.flavorblast.sbs\n;\n const LINEAR
= 0.8, HALF_BASE = 0.178, MAX_WHEEL = 0.9, RATE_MS = 100;\n\n // ANGULAR: start from localStorage or
default 4.0\n const LS_ANG = \nstarCockpit.angular\n;\n let ANGULAR =
parseFloat(localStorage.getItem(LS_ANG));\n if (!(ANGULAR > 0)) ANGULAR = 4.0;\n\n // DOM\n const img =
document.getElementById(\nstar-map-img\n);\n const status = document.getElementById(\nstar-status\n);\n
const swEstop = document.getElementById(\nstar-sw-estop\n);\n const swAuto =
document.getElementById(\nstar-sw-auto\n);\n const spInput = document.getElementById(\nstar-speed\n);\n
const spApply = document.getElementById(\nstar-speed-apply\n);\n const trInput =
document.getElementById(\nstar-turn\n);\n const trApply = document.getElementById(\nstar-turn-apply\n);\n
const btnRecord = document.getElementById(\nstar-map-record\n);\n const btnClear =
document.getElementById(\nstar-map-clear\n);\n const cockpit = document.getElementById(\nstar-cockpit\n);\n
const teleopGrid = document.getElementById(\nstar-teleop\n);\n\n trInput.value = ANGULAR;\n\n // --- state
render ---\n let _slamState = null;\n function render(state){\n if (state.estop === true) {\n
swEstop.classList.add(\non\n);\n swEstop.querySelector(\n.label\n).textContent = \nE-STOPPED\n;\n }\n else {\n
swEstop.classList.remove(\non\n);\n swEstop.querySelector(\n.label\n).textContent = \nE-STOP\n;\n }\n if
(state.autonomy === true) {\n swAuto.classList.add(\non\n);\n swAuto.querySelector(\n.label\n).textContent =
\nAUTONOMY\n;\n }\n else {\n swAuto.classList.remove(\non\n);\n swAuto.querySelector(\n.label\n).textContent =
\nMANUAL\n;\n }\n if (typeof state.speed === \nnumber\n) spInput.placeholder =
Math.round(state.speed*100);\n }\n function renderRecord(){\n if (_slamState === \nactive\n) {\n
btnRecord.textContent = \n[PAUSE] STOP MAPPING\n;\n btnRecord.style.background = \n#442\n;\n

```

```

1174         "mode": "html"
1175     },
1176     "fieldConfig": {
1177         "defaults": {},
1178         "overrides": []
1179     }
1180 }
1181 }
1182 }
1183 },
1184 "panel-3": {
1185     "kind": "Panel",
1186     "spec": {
1187         "id": 3,
1188         "title": "BL velocity",
1189         "description": "",
1190         "links": [],
1191         "data": {
1192             "kind": "QueryGroup",
1193             "spec": {
1194                 "queries": [
1195                     {
1196                         "kind": "PanelQuery",
1197                         "spec": {
1198                             "query": {
1199                                 "kind": "DataQuery",
1200                                 "group": "",
1201                                 "version": "v0",
1202                                 "spec": {
1203                                     "expr": "star_motor_velocity_mps{wheel=\"bl\"}",
1204                                     "legendFormat": ""
1205                                 }
1206                             },
1207                             "refId": "A",
1208                             "hidden": false
1209                         }
1210                     }
1211                 ],
1212                 "transformations": [],
1213                 "queryOptions": {}
1214             }
1215         },
1216         "vizConfig": {
1217             "kind": "VizConfig",
1218             "group": "gauge",
1219             "version": "",
1220             "spec": {
1221                 "options": {},
1222                 "fieldConfig": {
1223                     "defaults": {
1224                         "unit": "velocityms",
1225                         "min": -0.3,
1226                         "max": 0.3,
1227                         "thresholds": {
1228                             "mode": "absolute",
1229                             "steps": [
1230                                 {
1231                                     "value": null,
1232                                     "color": "blue"
1233                                 },
1234                                 {
1235                                     "value": -0.25,
1236                                     "color": "green"
1237                                 },
1238                                 {
1239                                     "value": 0.2,
1240                                     "color": "yellow"
1241                                 },
1242                                 {
1243                                     "value": 0.25,
1244                                     "color": "red"
1245                                 }
1246                             ]
1247                         }
1248                     },
1249                     "overrides": []
1250                 }
1251             }
1252         }
1253     }
1254 }

```

```

1253     }
1254 },
1255 "panel-4": {
1256   "kind": "Panel",
1257   "spec": {
1258     "id": 4,
1259     "title": "BR velocity",
1260     "description": "",
1261     "links": [],
1262     "data": {
1263       "kind": "QueryGroup",
1264       "spec": {
1265         "queries": [
1266           {
1267             "kind": "PanelQuery",
1268             "spec": {
1269               "query": {
1270                 "kind": "DataQuery",
1271                 "group": "",
1272                 "version": "v0",
1273                 "spec": {
1274                   "expr": "star_motor_velocity_mps{wheel=\"br\"}",
1275                   "legendFormat": ""
1276                 }
1277             },
1278             "refId": "A",
1279             "hidden": false
1280           }
1281         ],
1282         "transformations": [],
1283         "queryOptions": {}
1284       }
1285     },
1286   },
1287   "vizConfig": {
1288     "kind": "VizConfig",
1289     "group": "gauge",
1290     "version": "",
1291     "spec": {
1292       "options": {},
1293       "fieldConfig": {
1294         "defaults": {
1295           "unit": "velocityms",
1296           "min": -0.3,
1297           "max": 0.3,
1298           "thresholds": {
1299             "mode": "absolute",
1300             "steps": [
1301               {
1302                 "value": null,
1303                 "color": "blue"
1304               },
1305               {
1306                 "value": -0.25,
1307                 "color": "green"
1308               },
1309               {
1310                 "value": 0.2,
1311                 "color": "yellow"
1312               },
1313               {
1314                 "value": 0.25,
1315                 "color": "red"
1316               }
1317             ]
1318           }
1319         },
1320         "overrides": []
1321       }
1322     }
1323   },
1324 },
1325 "panel-5": {
1326   "kind": "Panel",
1327   "spec": {
1328     "id": 5,
1329     "title": "FL duty",
1330     "description": "",

```

```

1332     "links": [],
1333     "data": {
1334       "kind": "QueryGroup",
1335       "spec": {
1336         "queries": [
1337           {
1338             "kind": "PanelQuery",
1339             "spec": {
1340               "query": {
1341                 "kind": "DataQuery",
1342                 "group": "",
1343                 "version": "v0",
1344                 "spec": {
1345                   "expr": "star_motor_duty_percent{wheel=\"fl\"}"
1346                 }
1347             },
1348             "refId": "A",
1349             "hidden": false
1350           }
1351         ]
1352       },
1353       "transformations": [],
1354       "queryOptions": {}
1355     }
1356   },
1357   "vizConfig": {
1358     "kind": "VizConfig",
1359     "group": "gauge",
1360     "version": "",
1361     "spec": {
1362       "options": {},
1363       "fieldConfig": {
1364         "defaults": {
1365           "unit": "percent",
1366           "min": -100,
1367           "max": 100,
1368           "thresholds": {
1369             "mode": "absolute",
1370             "steps": [
1371               {
1372                 "value": null,
1373                 "color": "green"
1374               },
1375               {
1376                 "value": 60,
1377                 "color": "yellow"
1378               },
1379               {
1380                 "value": 85,
1381                 "color": "red"
1382               }
1383             ]
1384           }
1385         },
1386         "overrides": []
1387       }
1388     }
1389   }
1390 },
1391 "panel-6": {
1392   "kind": "Panel",
1393   "spec": {
1394     "id": 6,
1395     "title": "FR duty",
1396     "description": "",
1397     "links": [],
1398     "data": {
1399       "kind": "QueryGroup",
1400       "spec": {
1401         "queries": [
1402           {
1403             "kind": "PanelQuery",
1404             "spec": {
1405               "query": {
1406                 "kind": "DataQuery",
1407                 "group": "",
1408                 "version": "v0",
1409                 "spec": {

```

```

1411         "expr": "star_motor_duty_percent{wheel=\"fr\"}"
1412     },
1413 },
1414     "refId": "A",
1415     "hidden": false
1416 }
1417 },
1418 ],
1419     "transformations": [],
1420     "queryOptions": {}
1421 }
1422 },
1423     "vizConfig": {
1424         "kind": "VizConfig",
1425         "group": "gauge",
1426         "version": "",
1427         "spec": {
1428             "options": {},
1429             "fieldConfig": {
1430                 "defaults": {
1431                     "unit": "percent",
1432                     "min": -100,
1433                     "max": 100,
1434                     "thresholds": {
1435                         "mode": "absolute",
1436                         "steps": [
1437                             {
1438                                 "value": null,
1439                                 "color": "green"
1440                             },
1441                             {
1442                                 "value": 60,
1443                                 "color": "yellow"
1444                             },
1445                             {
1446                                 "value": 85,
1447                                 "color": "red"
1448                             }
1449                         ]
1450                     }
1451                 },
1452                 "overrides": []
1453             }
1454         }
1455     },
1456 },
1457 },
1458     "panel-7": {
1459         "kind": "Panel",
1460         "spec": {
1461             "id": 7,
1462             "title": "BL duty",
1463             "description": "",
1464             "links": [],
1465             "data": {
1466                 "kind": "QueryGroup",
1467                 "spec": {
1468                     "queries": [
1469                         {
1470                             "kind": "PanelQuery",
1471                             "spec": {
1472                                 "query": {
1473                                     "kind": "DataQuery",
1474                                     "group": "",
1475                                     "version": "v0",
1476                                     "spec": {
1477                                         "expr": "star_motor_duty_percent{wheel=\"bl\"}"
1478                                     }
1479                                 },
1480                                 "refId": "A",
1481                                 "hidden": false
1482                             }
1483                         }
1484                     ],
1485                     "transformations": [],
1486                     "queryOptions": {}
1487                 }
1488             },
1489             "vizConfig": {

```

```

1490     "kind": "VizConfig",
1491     "group": "gauge",
1492     "version": "",
1493     "spec": {
1494       "options": {},
1495       "fieldConfig": {
1496         "defaults": {
1497           "unit": "percent",
1498           "min": -100,
1499           "max": 100,
1500           "thresholds": {
1501             "mode": "absolute",
1502             "steps": [
1503               {
1504                 "value": null,
1505                 "color": "green"
1506               },
1507               {
1508                 "value": 60,
1509                 "color": "yellow"
1510               },
1511               {
1512                 "value": 85,
1513                 "color": "red"
1514               }
1515             ]
1516           }
1517         },
1518         "overrides": []
1519       }
1520     }
1521   },
1522 },
1523 {
1524   "panel-8": {
1525     "kind": "Panel",
1526     "spec": {
1527       "id": 8,
1528       "title": "BR duty",
1529       "description": "",
1530       "links": [],
1531       "data": {
1532         "kind": "QueryGroup",
1533         "spec": {
1534           "queries": [
1535             {
1536               "kind": "PanelQuery",
1537               "spec": {
1538                 "query": {
1539                   "kind": "DataQuery",
1540                   "group": "",
1541                   "version": "v0",
1542                   "spec": {
1543                     "expr": "star_motor_duty_percent{wheel=\"br\"}"
1544                   }
1545                 },
1546                 "refId": "A",
1547                 "hidden": false
1548               }
1549             }
1550           ],
1551           "transformations": [],
1552           "queryOptions": {}
1553         }
1554       },
1555       "vizConfig": {
1556         "kind": "VizConfig",
1557         "group": "gauge",
1558         "version": "",
1559         "spec": {
1560           "options": {},
1561           "fieldConfig": {
1562             "defaults": {
1563               "unit": "percent",
1564               "min": -100,
1565               "max": 100,
1566               "thresholds": {
1567                 "mode": "absolute",
1568                 "steps": [

```

```

1569         {
1570             "value": null,
1571             "color": "green"
1572         },
1573         {
1574             "value": 60,
1575             "color": "yellow"
1576         },
1577         {
1578             "value": 85,
1579             "color": "red"
1580         }
1581     ]
1582 }
1583 },
1584 "overrides": []
1585 }
1586 }
1587 }
1588 },
1589 "panel-9": {
1590     "kind": "Panel",
1591     "spec": {
1592         "id": 9,
1593         "title": "Wheel velocities over time",
1594         "description": "",
1595         "links": [],
1596         "data": {
1597             "kind": "QueryGroup",
1598             "spec": {
1599                 "queries": [
1600                     {
1601                         "kind": "PanelQuery",
1602                         "spec": {
1603                             "query": {
1604                                 "kind": "DataQuery",
1605                                 "group": "",
1606                                 "version": "v0",
1607                                 "spec": {
1608                                     "expr": "star_motor_velocity_mps",
1609                                     "legendFormat": "{{wheel}}"
1610                                 }
1611                             },
1612                             "refId": "A",
1613                             "hidden": false
1614                         }
1615                     }
1616                 ],
1617                 "transformations": [],
1618                 "queryOptions": {}
1619             }
1620         },
1621     },
1622     "vizConfig": {
1623         "kind": "VizConfig",
1624         "group": "timeseries",
1625         "version": "",
1626         "spec": {
1627             "options": {
1628                 "legend": {
1629                     "displayMode": "list",
1630                     "placement": "bottom",
1631                     "showLegend": true
1632                 }
1633             },
1634             "fieldConfig": {
1635                 "defaults": {
1636                     "unit": "velocityms"
1637                 },
1638                 "overrides": []
1639             }
1640         }
1641     }
1642 }
1643 },
1644 "layout": {
1645     "kind": "RowsLayout",
1646     "spec": {

```



```

1648 "rows": [
1649   {
1650     "kind": "RowsLayoutRow",
1651     "spec": {
1652       "title": "SLAM map",
1653       "collapse": false,
1654       "layout": {
1655         "kind": "GridLayout",
1656         "spec": {
1657           "items": [
1658             {
1659               "kind": "GridLayoutItem",
1660               "spec": {
1661                 "x": 0,
1662                 "y": 0,
1663                 "width": 24,
1664                 "height": 22,
1665                 "element": {
1666                   "kind": "ElementReference",
1667                   "name": "panel-29"
1668                 }
1669             }
1670           ],
1671           {
1672             "kind": "GridLayoutItem",
1673             "spec": {
1674               "x": 0,
1675               "y": 0,
1676               "width": 6,
1677               "height": 7,
1678               "element": {
1679                 "kind": "ElementReference",
1680                 "name": "panel-1"
1681               }
1682             }
1683           },
1684           {
1685             "kind": "GridLayoutItem",
1686             "spec": {
1687               "x": 6,
1688               "y": 0,
1689               "width": 6,
1690               "height": 7,
1691               "element": {
1692                 "kind": "ElementReference",
1693                 "name": "panel-2"
1694               }
1695             }
1696           },
1697           {
1698             "kind": "GridLayoutItem",
1699             "spec": {
1700               "x": 12,
1701               "y": 0,
1702               "width": 6,
1703               "height": 7,
1704               "element": {
1705                 "kind": "ElementReference",
1706                 "name": "panel-3"
1707               }
1708             }
1709           },
1710           {
1711             "kind": "GridLayoutItem",
1712             "spec": {
1713               "x": 18,
1714               "y": 0,
1715               "width": 6,
1716               "height": 7,
1717               "element": {
1718                 "kind": "ElementReference",
1719                 "name": "panel-4"
1720               }
1721             }
1722           },
1723           {
1724             "kind": "GridLayoutItem",
1725             "spec": {
1726               "x": 0,

```

```

1727         "y": 7,
1728         "width": 6,
1729         "height": 7,
1730         "element": {
1731             "kind": "ElementReference",
1732             "name": "panel-5"
1733         }
1734     },
1735 },
1736 {
1737     "kind": "GridLayoutItem",
1738     "spec": {
1739         "x": 6,
1740         "y": 7,
1741         "width": 6,
1742         "height": 7,
1743         "element": {
1744             "kind": "ElementReference",
1745             "name": "panel-6"
1746         }
1747     }
1748 },
1749 {
1750     "kind": "GridLayoutItem",
1751     "spec": {
1752         "x": 12,
1753         "y": 7,
1754         "width": 6,
1755         "height": 7,
1756         "element": {
1757             "kind": "ElementReference",
1758             "name": "panel-7"
1759         }
1760     }
1761 },
1762 {
1763     "kind": "GridLayoutItem",
1764     "spec": {
1765         "x": 18,
1766         "y": 7,
1767         "width": 6,
1768         "height": 7,
1769         "element": {
1770             "kind": "ElementReference",
1771             "name": "panel-8"
1772         }
1773     }
1774 }
1775 ]
1776 }
1777 }
1778 },
1779 {
1780     "kind": "RowLayoutRow",
1781     "spec": {
1782         "title": "Motors",
1783         "collapse": false,
1784         "layout": {
1785             "kind": "GridLayout",
1786             "spec": {
1787                 "items": [
1788                     {
1789                         "kind": "GridLayoutItem",
1790                         "spec": {
1791                             "x": 0,
1792                             "y": -1,
1793                             "width": 12,
1794                             "height": 8,
1795                             "element": {
1796                                 "kind": "ElementReference",
1797                                 "name": "panel-9"
1798                             }
1799                         }
1800                     }
1801                 ]
1802             }
1803             "kind": "GridLayoutItem",
1804             "spec": {
1805                 "x": 12,

```

```

1806         "y": -1,
1807         "width": 12,
1808         "height": 8,
1809         "element": {
1810             "kind": "ElementReference",
1811             "name": "panel-10"
1812         }
1813     },
1814 },
1815 {
1816     "kind": "GridLayoutItem",
1817     "spec": {
1818         "x": 0,
1819         "y": 7,
1820         "width": 24,
1821         "height": 4,
1822         "element": {
1823             "kind": "ElementReference",
1824             "name": "panel-11"
1825         }
1826     }
1827 },
1828 ]
1829 }
1830 }
1831 },
1832 },
1833 {
1834     "kind": "RowLayoutRow",
1835     "spec": {
1836         "title": "IMU",
1837         "collapse": false,
1838         "layout": {
1839             "kind": "GridLayout",
1840             "spec": {
1841                 "items": [
1842                     {
1843                         "kind": "GridLayoutItem",
1844                         "spec": {
1845                             "x": 0,
1846                             "y": 0,
1847                             "width": 8,
1848                             "height": 7,
1849                             "element": {
1850                                 "kind": "ElementReference",
1851                                 "name": "panel-12"
1852                             }
1853                         }
1854                     },
1855                     {
1856                         "kind": "GridLayoutItem",
1857                         "spec": {
1858                             "x": 8,
1859                             "y": 0,
1860                             "width": 8,
1861                             "height": 7,
1862                             "element": {
1863                                 "kind": "ElementReference",
1864                                 "name": "panel-13"
1865                             }
1866                         }
1867                     },
1868                     {
1869                         "kind": "GridLayoutItem",
1870                         "spec": {
1871                             "x": 16,
1872                             "y": 0,
1873                             "width": 8,
1874                             "height": 7,
1875                             "element": {
1876                                 "kind": "ElementReference",
1877                                 "name": "panel-14"
1878                             }
1879                         }
1880                     }
1881                 ],
1882                 "kind": "GridLayoutItem",
1883                 "spec": {
1884                     "x": 0,

```

```

1885         "y": 7,
1886         "width": 12,
1887         "height": 7,
1888         "element": {
1889             "kind": "ElementReference",
1890             "name": "panel-15"
1891         }
1892     },
1893 },
1894 {
1895     "kind": "GridLayoutItem",
1896     "spec": {
1897         "x": 12,
1898         "y": 7,
1899         "width": 12,
1900         "height": 7,
1901         "element": {
1902             "kind": "ElementReference",
1903             "name": "panel-16"
1904         }
1905     }
1906 }
1907 ]
1908 }
1909 }
1910 }
1911 },
1912 {
1913     "kind": "RowLayoutRow",
1914     "spec": {
1915         "title": "Odometry",
1916         "collapse": false,
1917         "layout": {
1918             "kind": "GridLayout",
1919             "spec": {
1920                 "items": [
1921                     {
1922                         "kind": "GridLayoutItem",
1923                         "spec": {
1924                             "x": 0,
1925                             "y": 0,
1926                             "width": 8,
1927                             "height": 6,
1928                             "element": {
1929                                 "kind": "ElementReference",
1930                                 "name": "panel-17"
1931                             }
1932                         }
1933                     },
1934                     {
1935                         "kind": "GridLayoutItem",
1936                         "spec": {
1937                             "x": 8,
1938                             "y": 0,
1939                             "width": 8,
1940                             "height": 6,
1941                             "element": {
1942                                 "kind": "ElementReference",
1943                                 "name": "panel-18"
1944                             }
1945                         }
1946                     },
1947                     {
1948                         "kind": "GridLayoutItem",
1949                         "spec": {
1950                             "x": 16,
1951                             "y": 0,
1952                             "width": 8,
1953                             "height": 6,
1954                             "element": {
1955                                 "kind": "ElementReference",
1956                                 "name": "panel-19"
1957                             }
1958                         }
1959                     }
1960                 ]
1961             }
1962         }
1963     }

```

```

},
{
  "kind": "RowLayoutRow",
  "spec": {
    "title": "LiDAR",
    "collapse": false,
    "layout": {
      "kind": "GridLayout",
      "spec": {
        "items": [
          {
            "kind": "GridLayoutItem",
            "spec": {
              "x": 0,
              "y": 0,
              "width": 6,
              "height": 6,
              "element": {
                "kind": "ElementReference",
                "name": "panel-24"
              }
            }
          },
          {
            "kind": "GridLayoutItem",
            "spec": {
              "x": 6,
              "y": 0,
              "width": 6,
              "height": 6,
              "element": {
                "kind": "ElementReference",
                "name": "panel-25"
              }
            }
          }
        ]
      }
    },
    {
      "kind": "GridLayoutItem",
      "spec": {
        "x": 12,
        "y": 0,
        "width": 6,
        "height": 6,
        "element": {
          "kind": "ElementReference",
          "name": "panel-26"
        }
      }
    }
  ],
  {
    "kind": "GridLayoutItem",
    "spec": {
      "x": 18,
      "y": 0,
      "width": 6,
      "height": 6,
      "element": {
        "kind": "ElementReference",
        "name": "panel-27"
      }
    }
  ]
}
},
{
  "kind": "RowLayoutRow",
  "spec": {
    "title": "Bridge health",
    "collapse": true,
    "layout": {
      "kind": "GridLayout",
      "spec": {
        "items": [
          {
            "kind": "GridLayoutItem",
            "spec": {

```

```

        "x": 0,
        "y": -15,
        "width": 12,
        "height": 8,
        "element": {
            "kind": "ElementReference",
            "name": "panel-20"
        }
    },
    {
        "kind": "GridLayoutItem",
        "spec": {
            "x": 12,
            "y": -15,
            "width": 12,
            "height": 8,
            "element": {
                "kind": "ElementReference",
                "name": "panel-21"
            }
        }
    },
    {
        "kind": "GridLayoutItem",
        "spec": {
            "x": 0,
            "y": -7,
            "width": 12,
            "height": 4,
            "element": {
                "kind": "ElementReference",
                "name": "panel-22"
            }
        }
    },
    {
        "kind": "GridLayoutItem",
        "spec": {
            "x": 12,
            "y": -7,
            "width": 12,
            "height": 4,
            "element": {
                "kind": "ElementReference",
                "name": "panel-23"
            }
        }
    }
]
}
},
"links": [],
"liveNow": false,
"preload": false,
"tags": [
    "star",
    "robot"
],
"timeSettings": {
    "from": "now-2m",
    "to": "now",
    "autoRefresh": "1s",
    "autoRefreshIntervals": [
        "5s",
        "10s",
        "30s",
        "1m",
        "5m",
        "15m",
        "30m",
        "1h",
        "2h",
        "1d"
    ]
}

```

```
2122     ],
2123     "hideTimepicker": false,
2124     "fiscalYearStartMonth": 0
2125   },
2126   "title": "STAR robot telemetry",
2127   "variables": []
2128 }
2129 }
```

STAR Grafana Dashboard

Rendered Snapshot

*Visual representations of every panel in the
monitoring/grafana-star-dashboard.json dashboard.*

Generated by `scripts/render-grafana-json.py`
using offline PIL rendering with synthetic demo data.

To capture screenshots of the production dashboard with live
Prometheus data, deploy via `monitoring/docker-compose.yml`
and use the grafana-image-renderer plugin.

Dashboard Overview

STAR robot telemetry

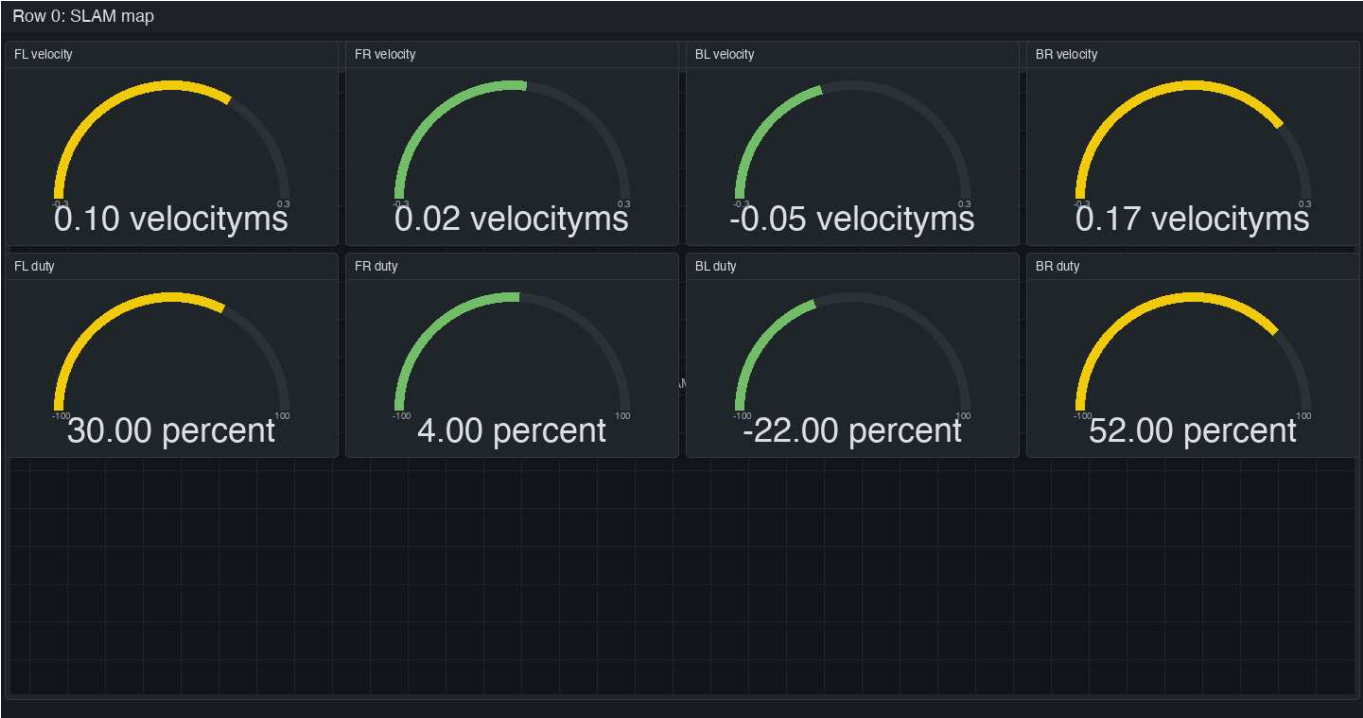
Composite of all rows



Per-Row Composites

Row 0: SLAM map

All panels in this row, arranged on the Grafana 24-column grid



Row 1: Motors

All panels in this row, arranged on the Grafana 24-column grid



Row 2: IMU

All panels in this row, arranged on the Grafana 24-column grid



Row 3: Odometry

All panels in this row, arranged on the Grafana 24-column grid



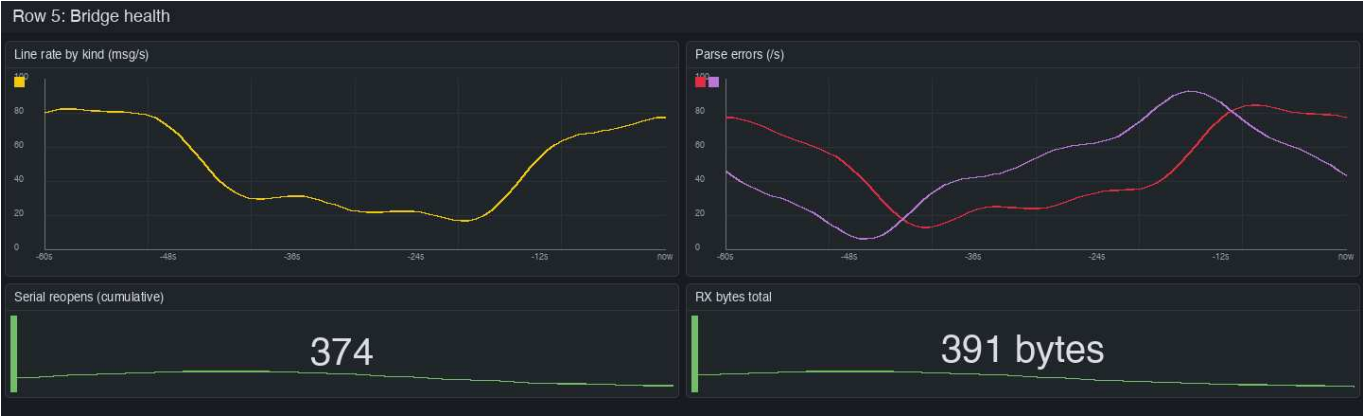
Row 4: LiDAR

All panels in this row, arranged on the Grafana 24-column grid



Row 5: Bridge health

All panels in this row, arranged on the Grafana 24-column grid

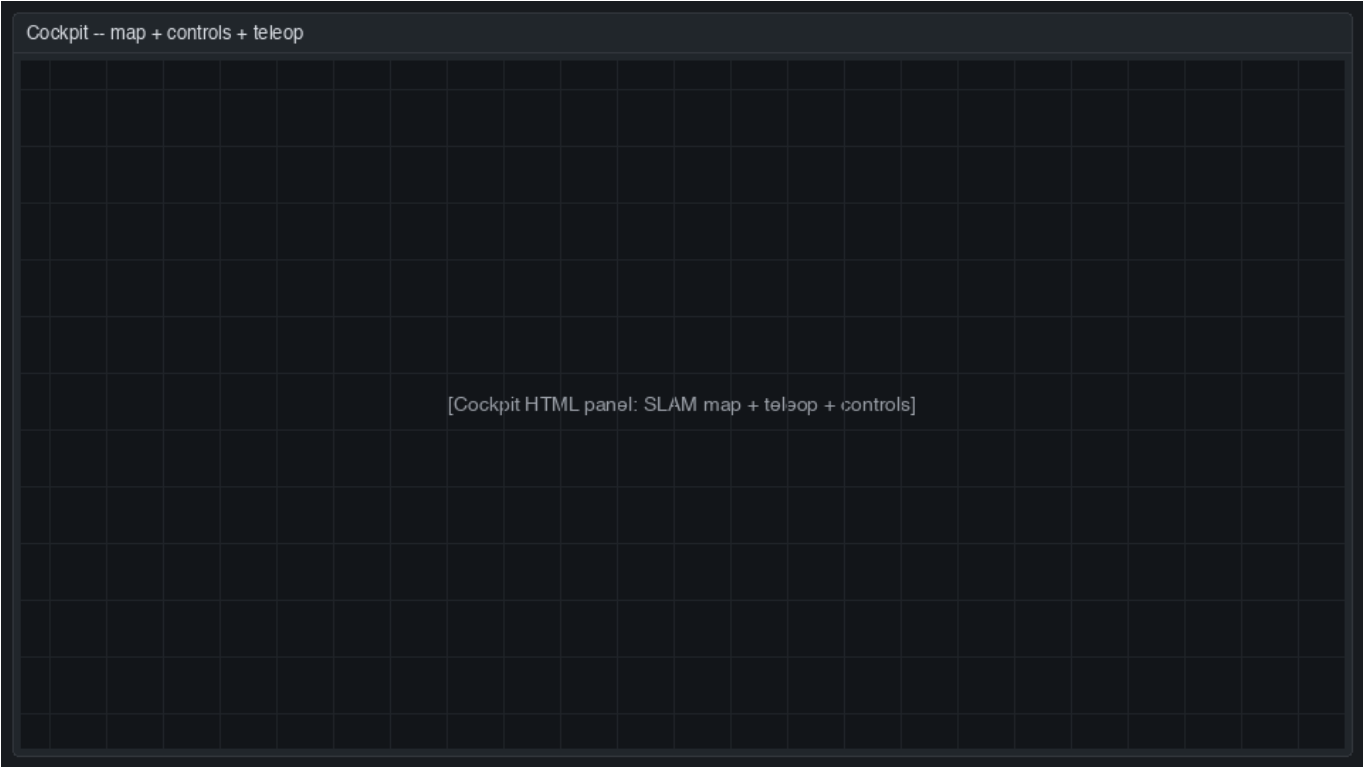


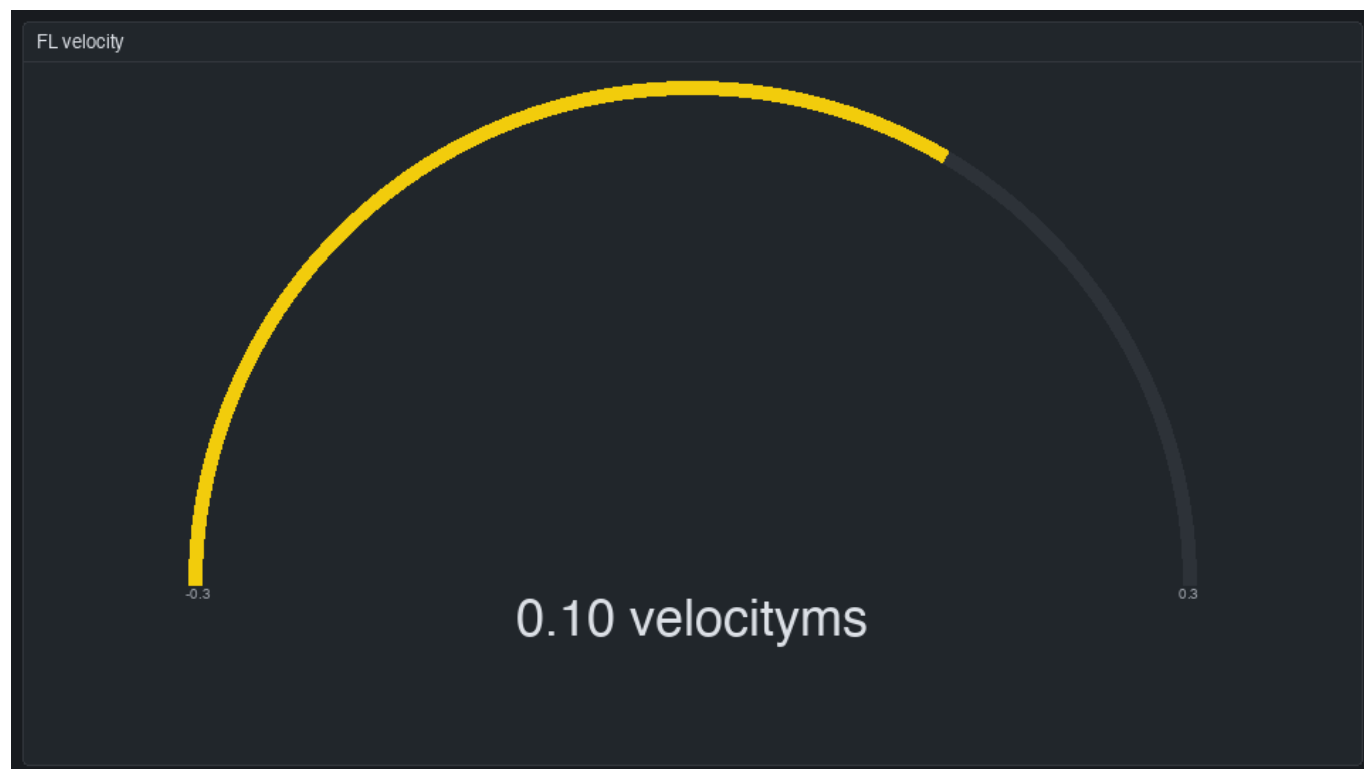
Individual Panels

Row 0: SLAM map

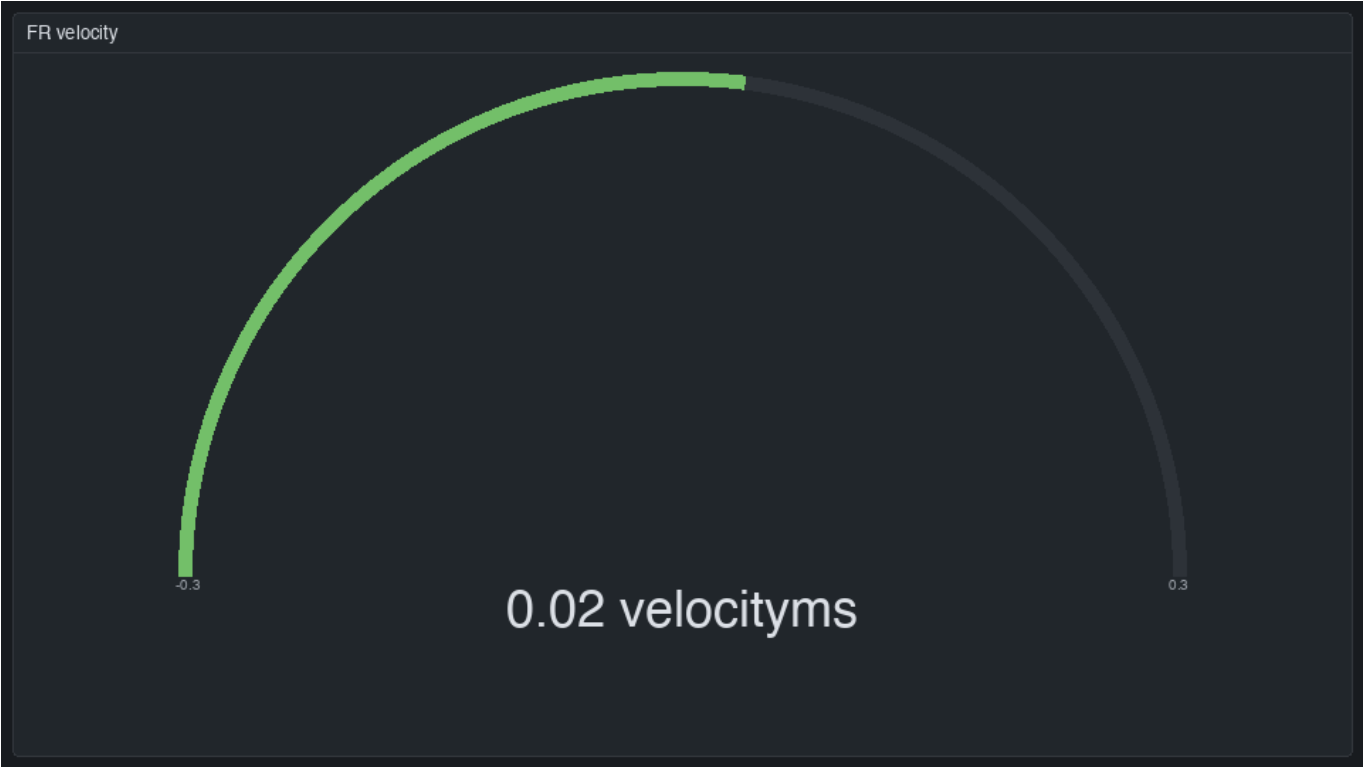
Cockpit – map + controls + teleop

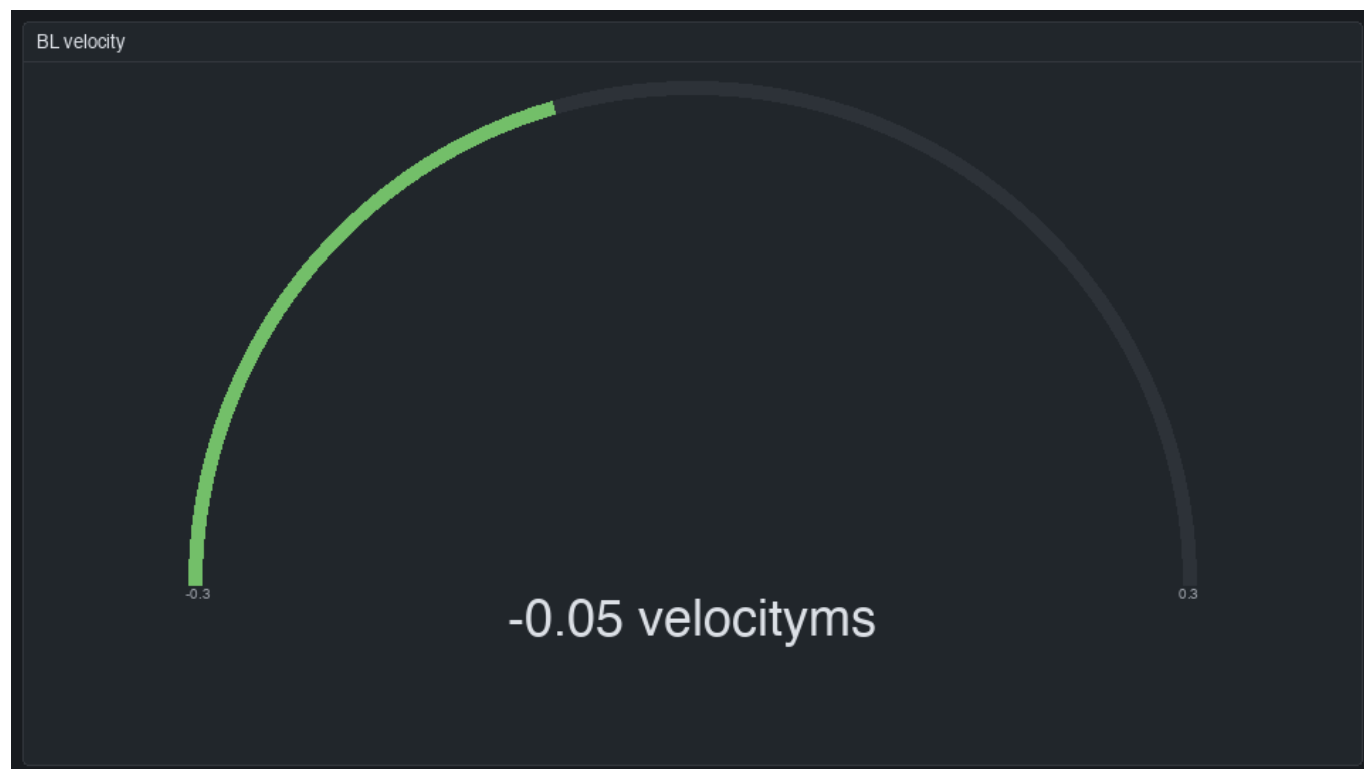
Row 0: SLAM map

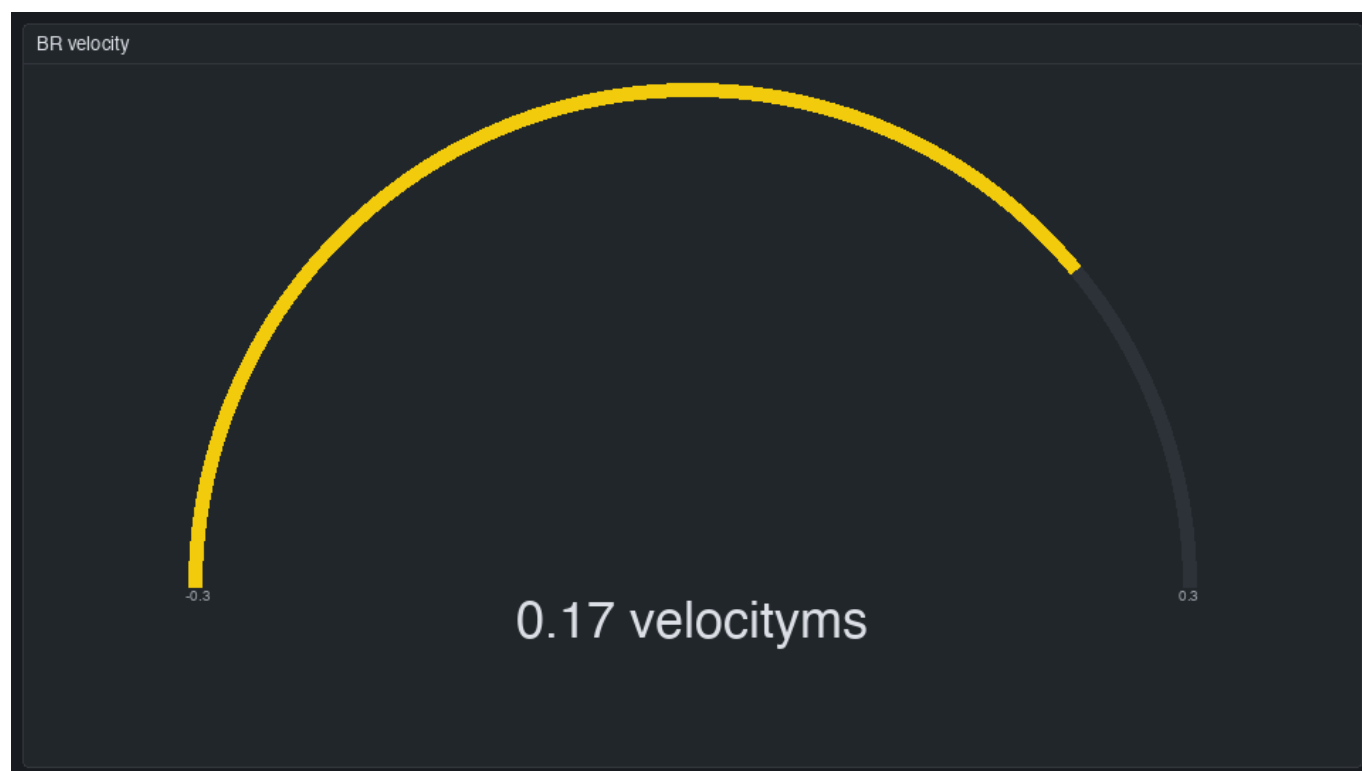


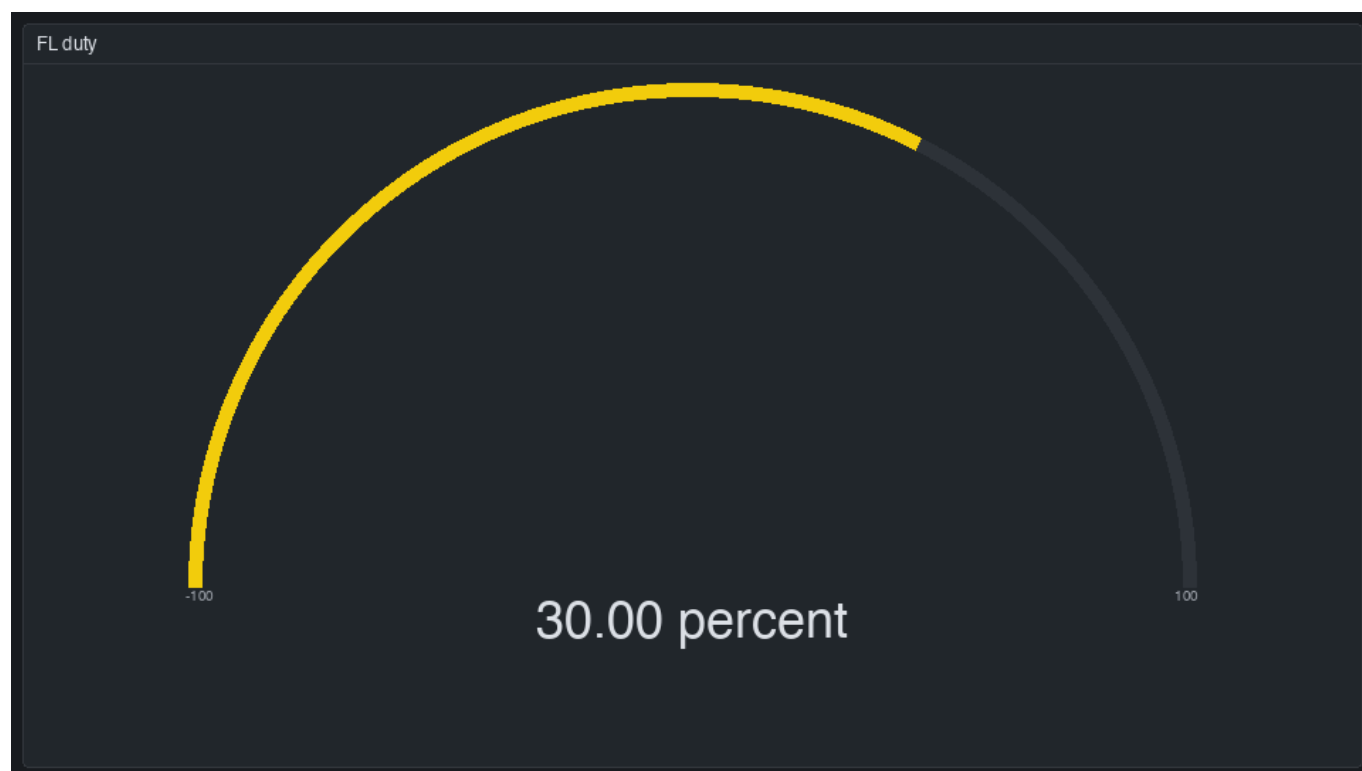
FL velocity*Row 0: SLAM map*

FR velocity
Row 0: SLAM map



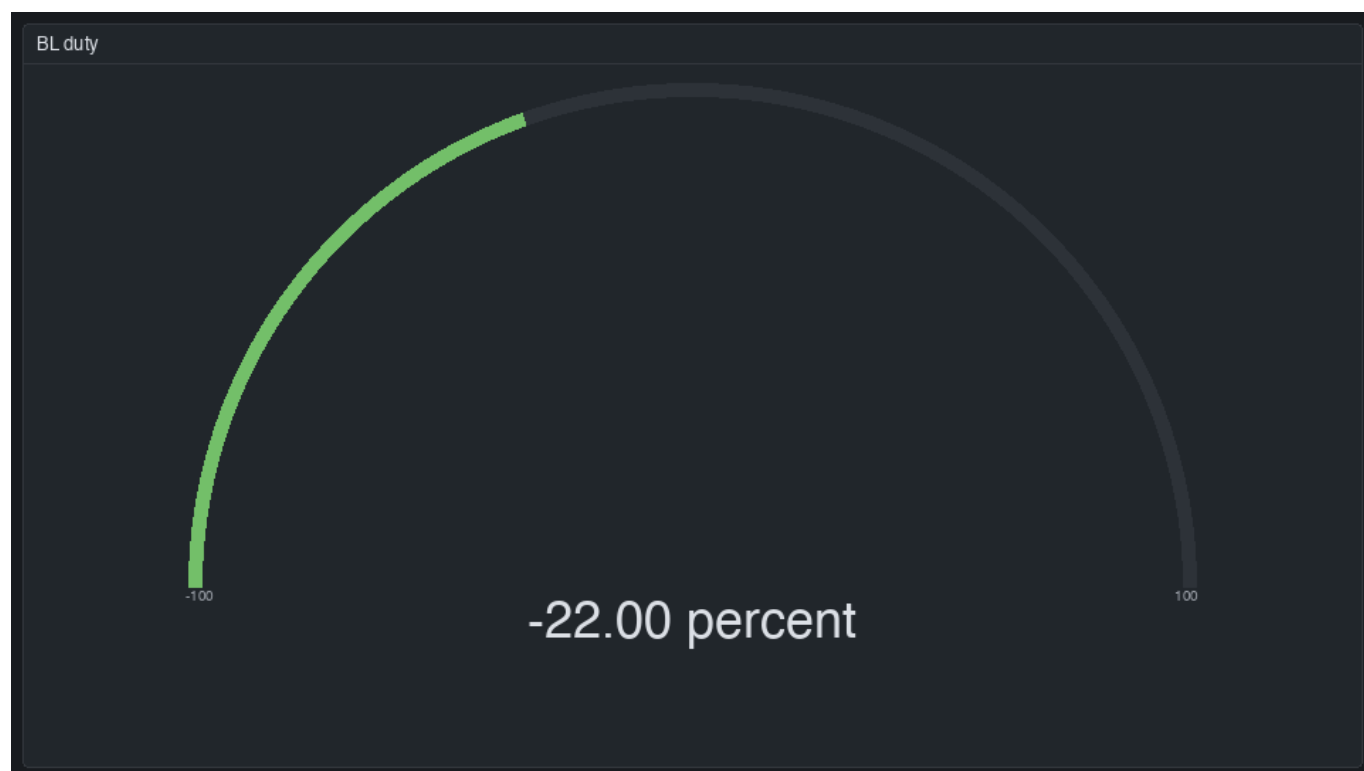
BL velocity*Row 0: SLAM map*

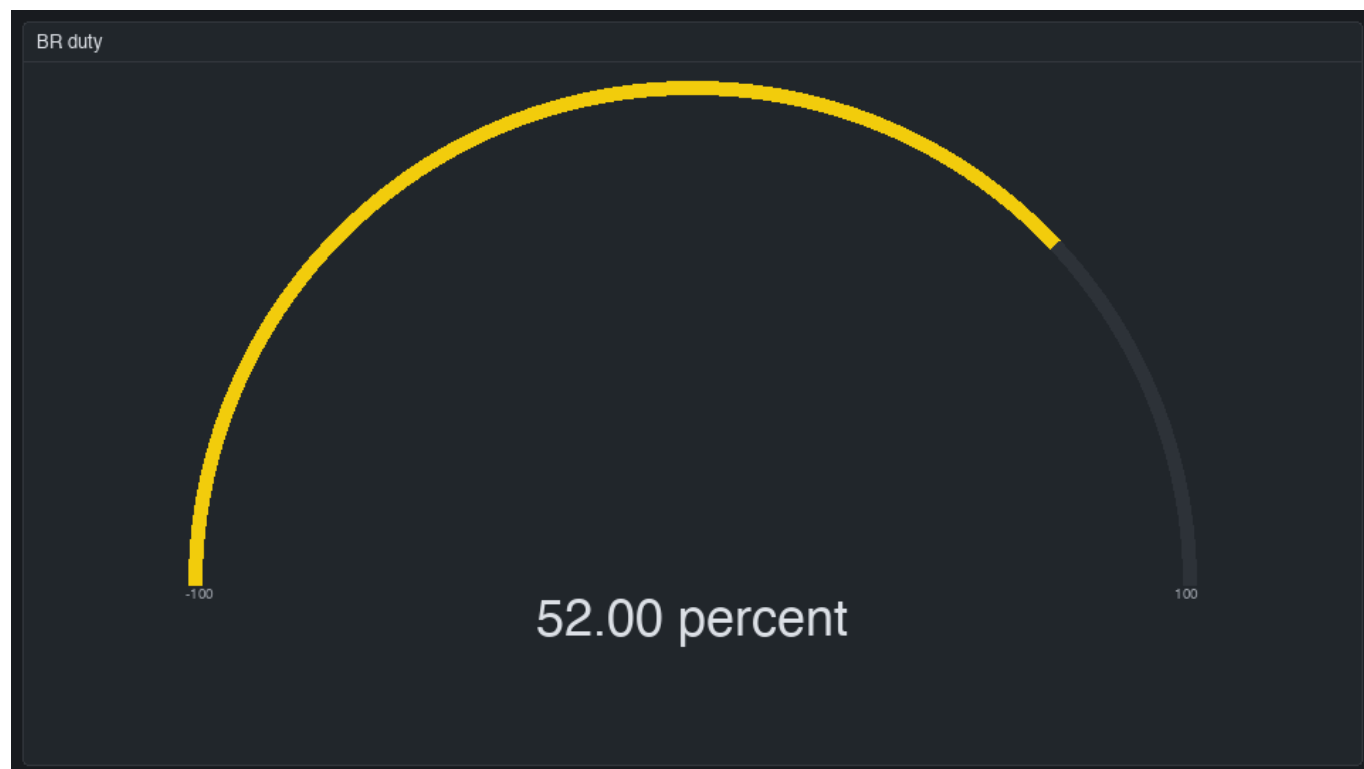
BR velocity*Row 0: SLAM map*

FL duty*Row 0: SLAM map*

FR duty
Row 0: SLAM map



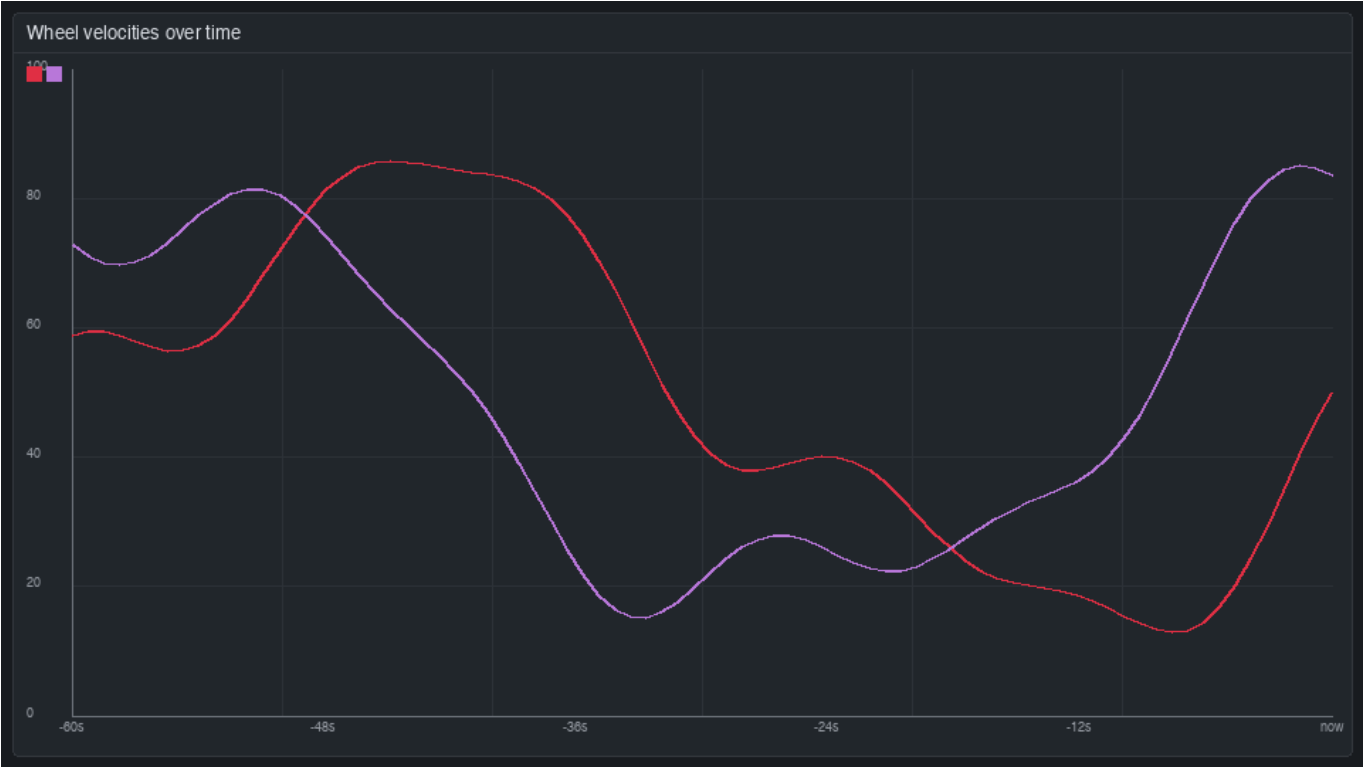
BL duty*Row 0: SLAM map*

BR duty*Row 0: SLAM map*

Row 1: Motors

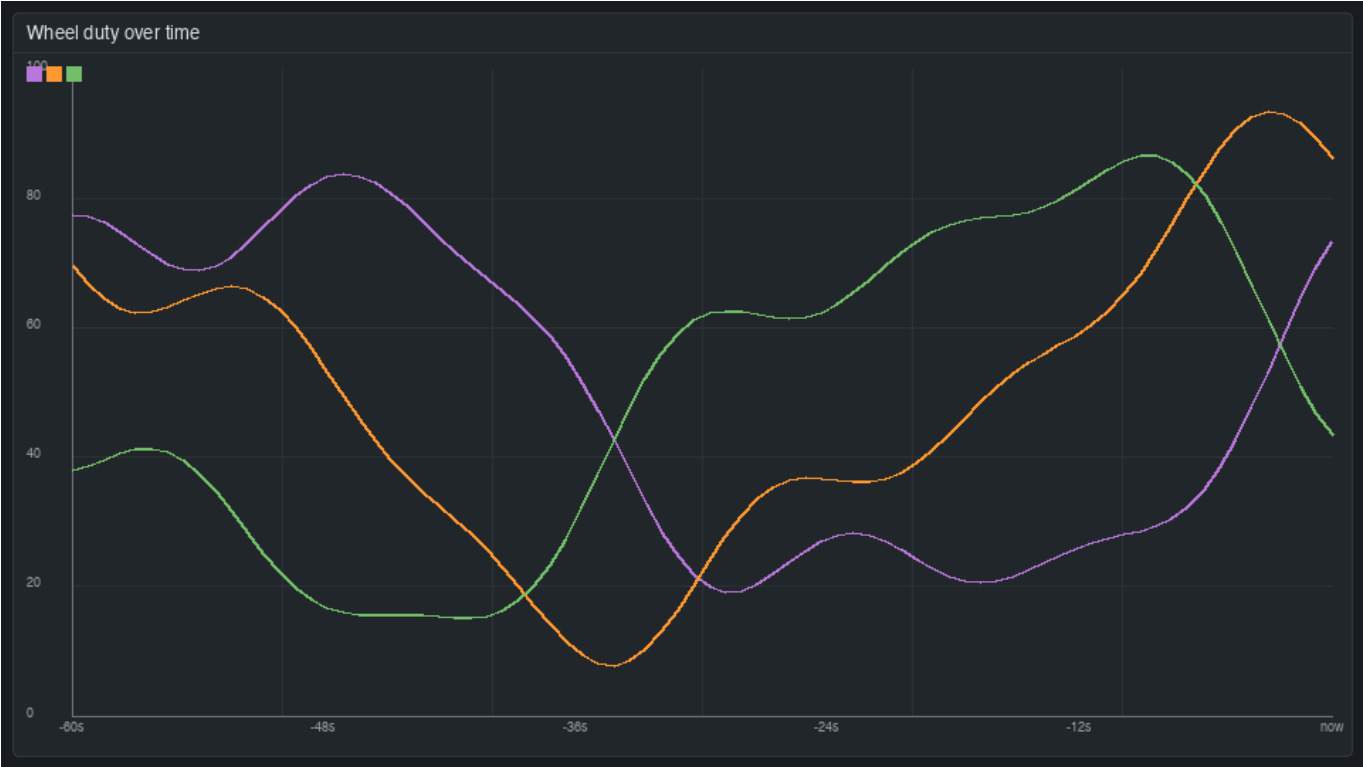
Wheel velocities over time

Row 1: Motors



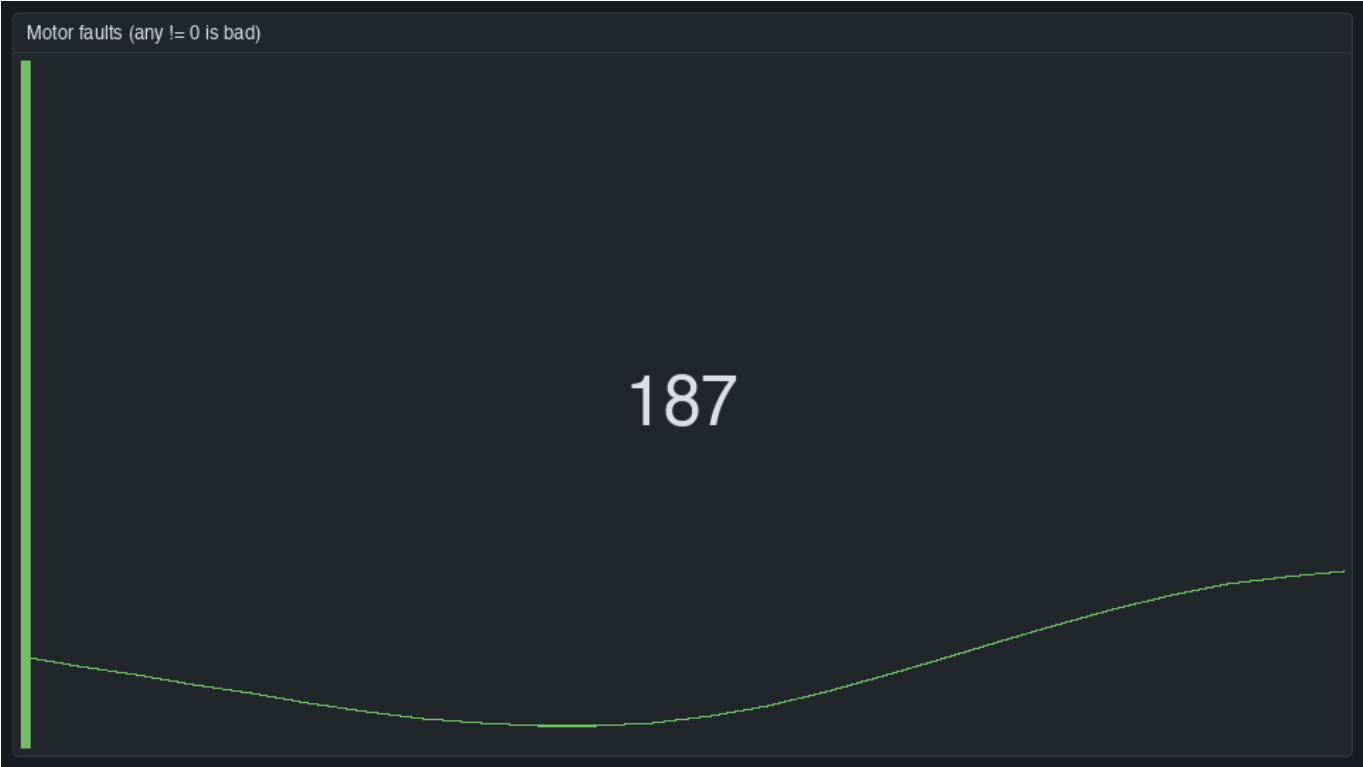
Wheel duty over time

Row 1: Motors



Motor faults (any != 0 is bad)

Row 1: Motors



Row 2: IMU

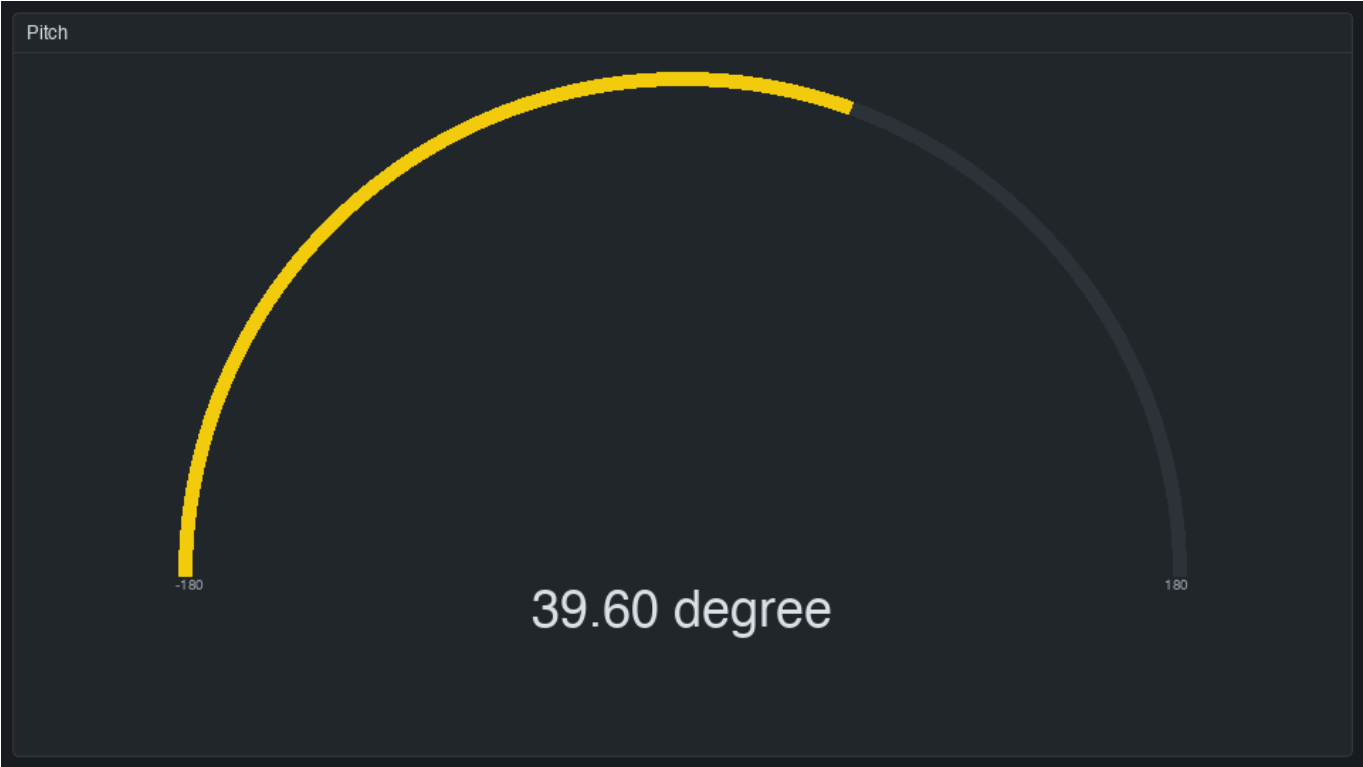
Roll

Row 2: IMU



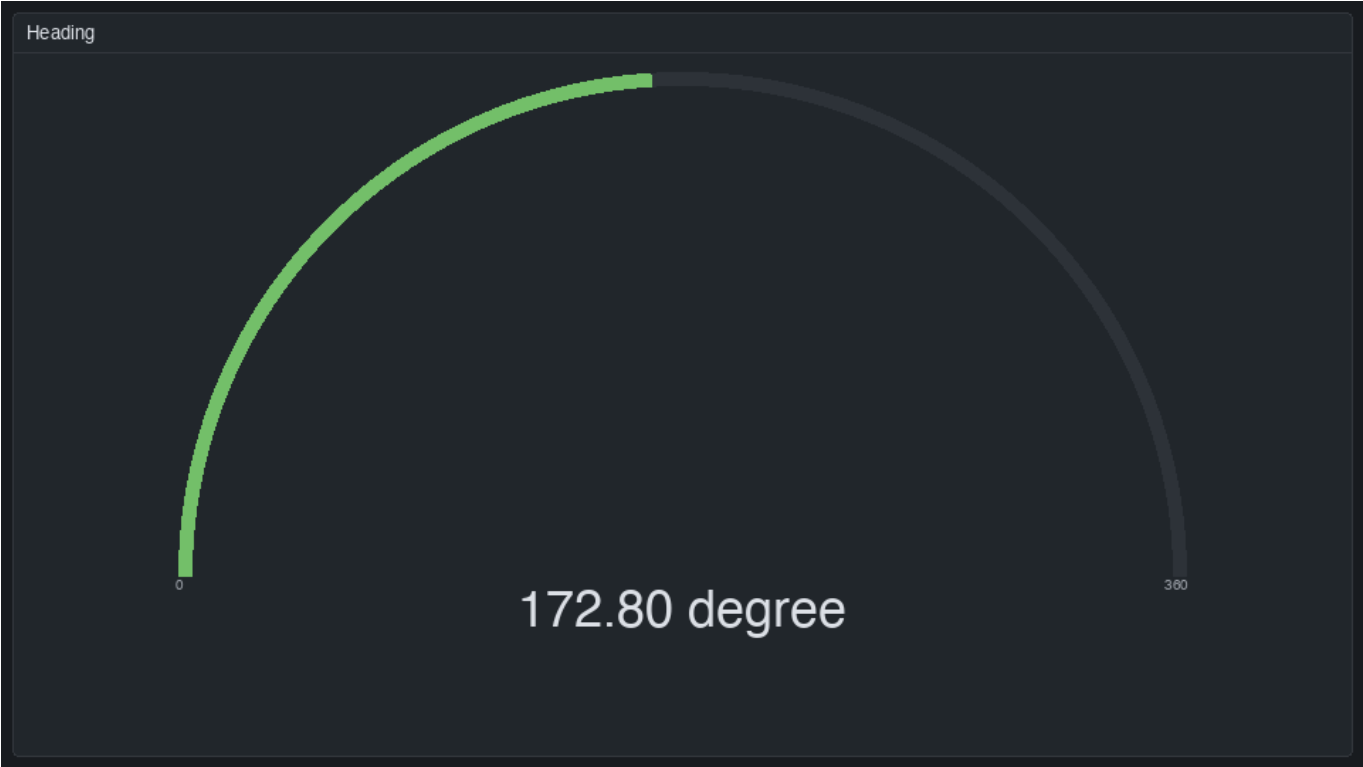
Pitch

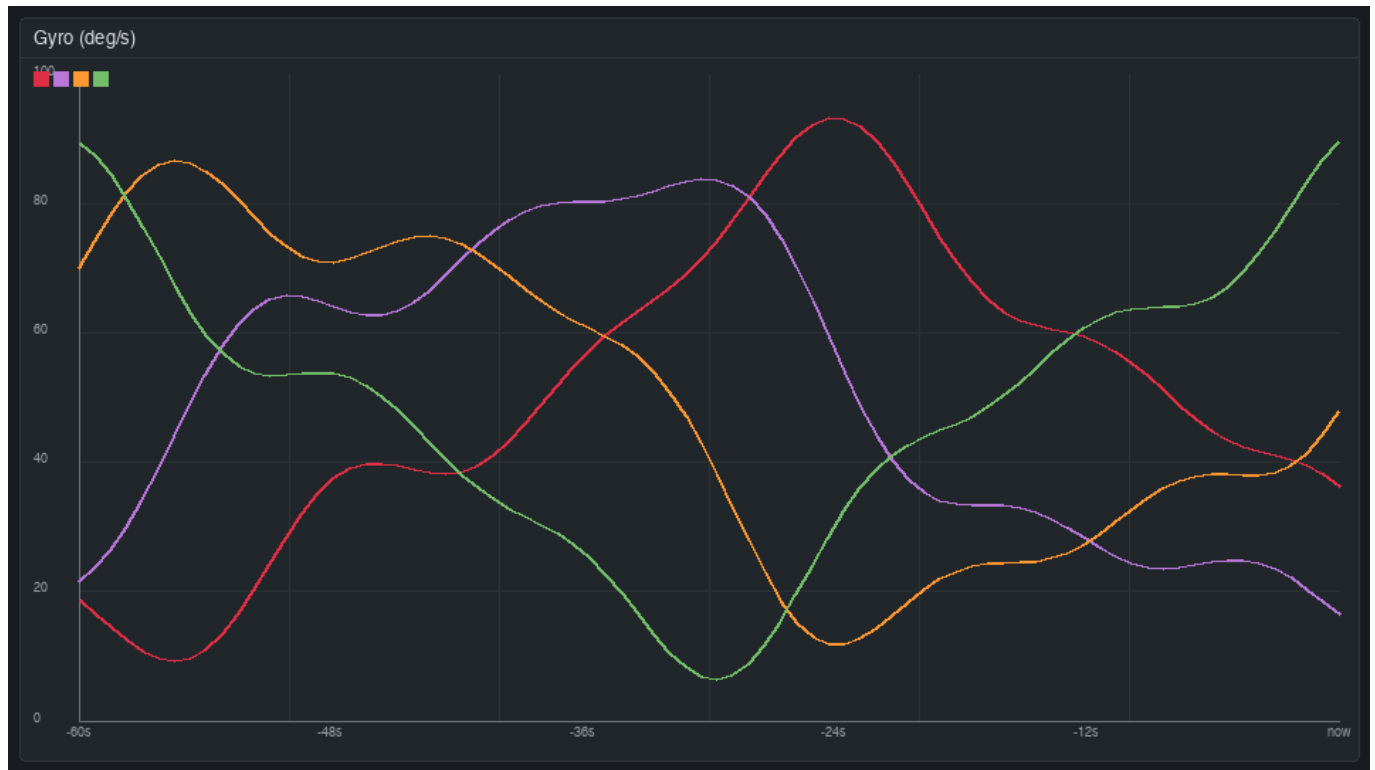
Row 2: IMU



Heading

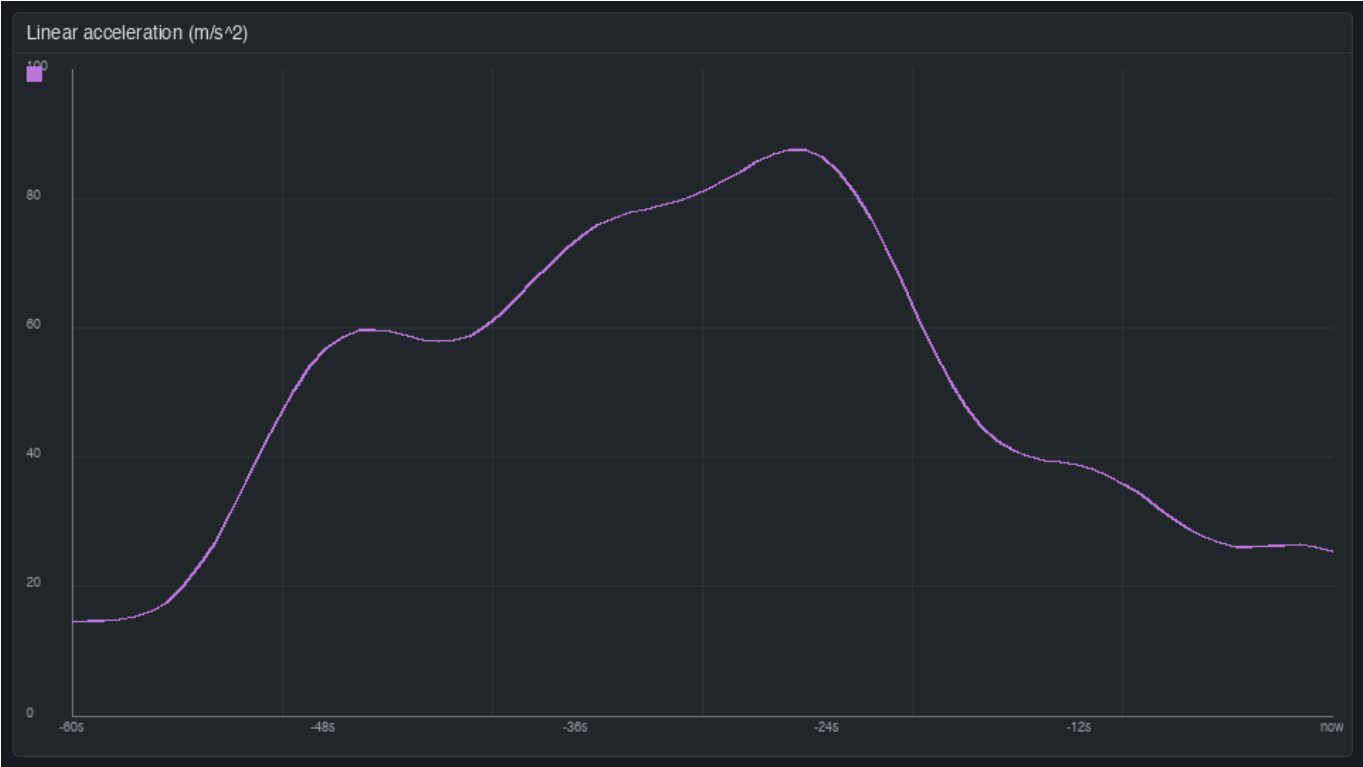
Row 2: IMU



Gyro (deg/s)*Row 2: IMU*

Linear acceleration (m/s²)

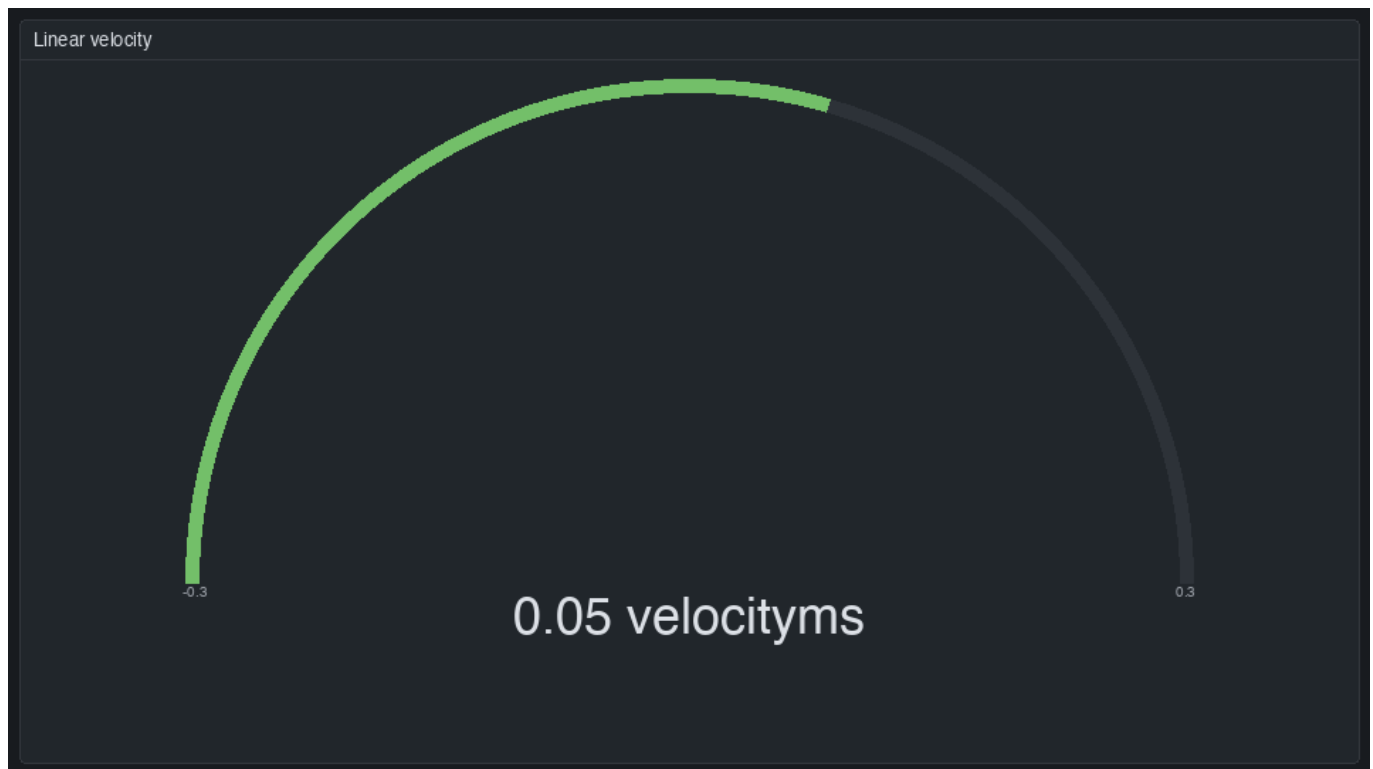
Row 2: IMU



Row 3: Odometry

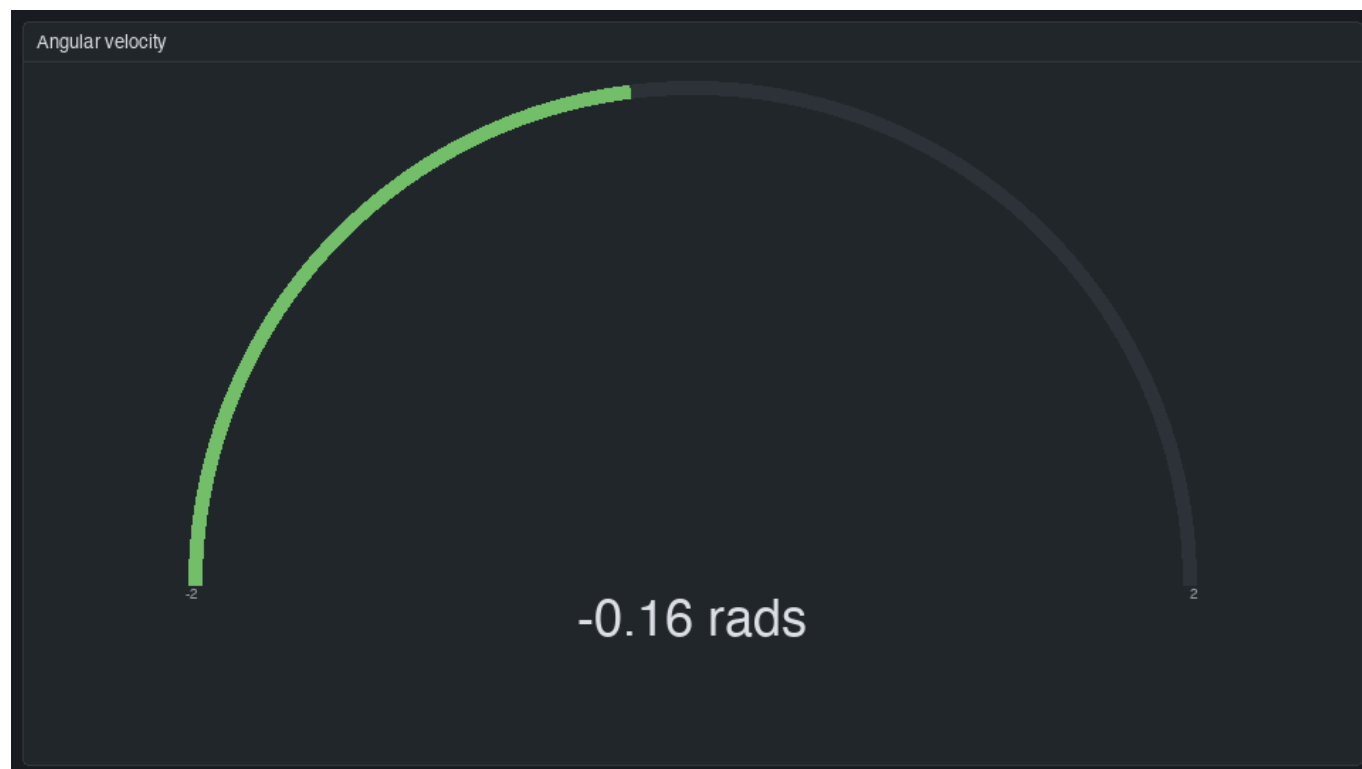
Linear velocity

Row 3: Odometry



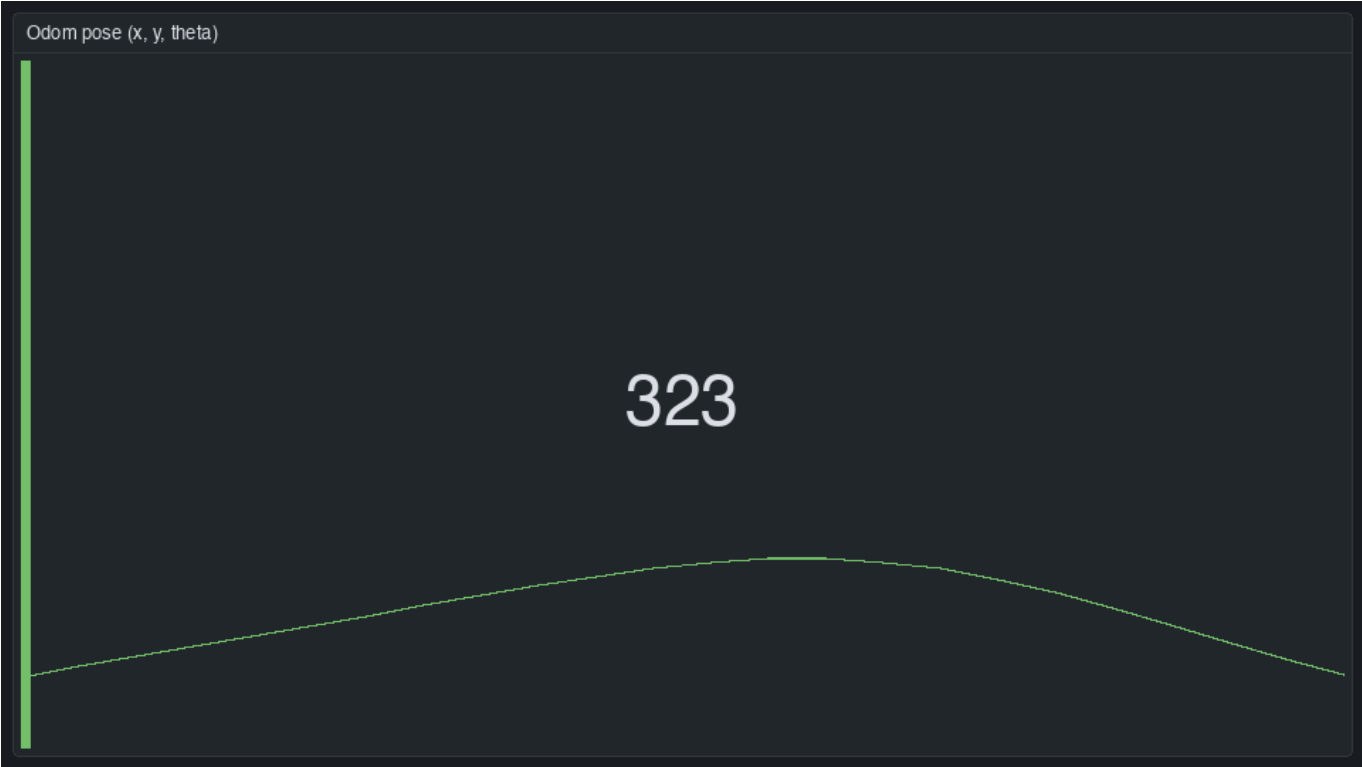
Angular velocity

Row 3: Odometry



Odom pose (x, y, theta)

Row 3: Odometry

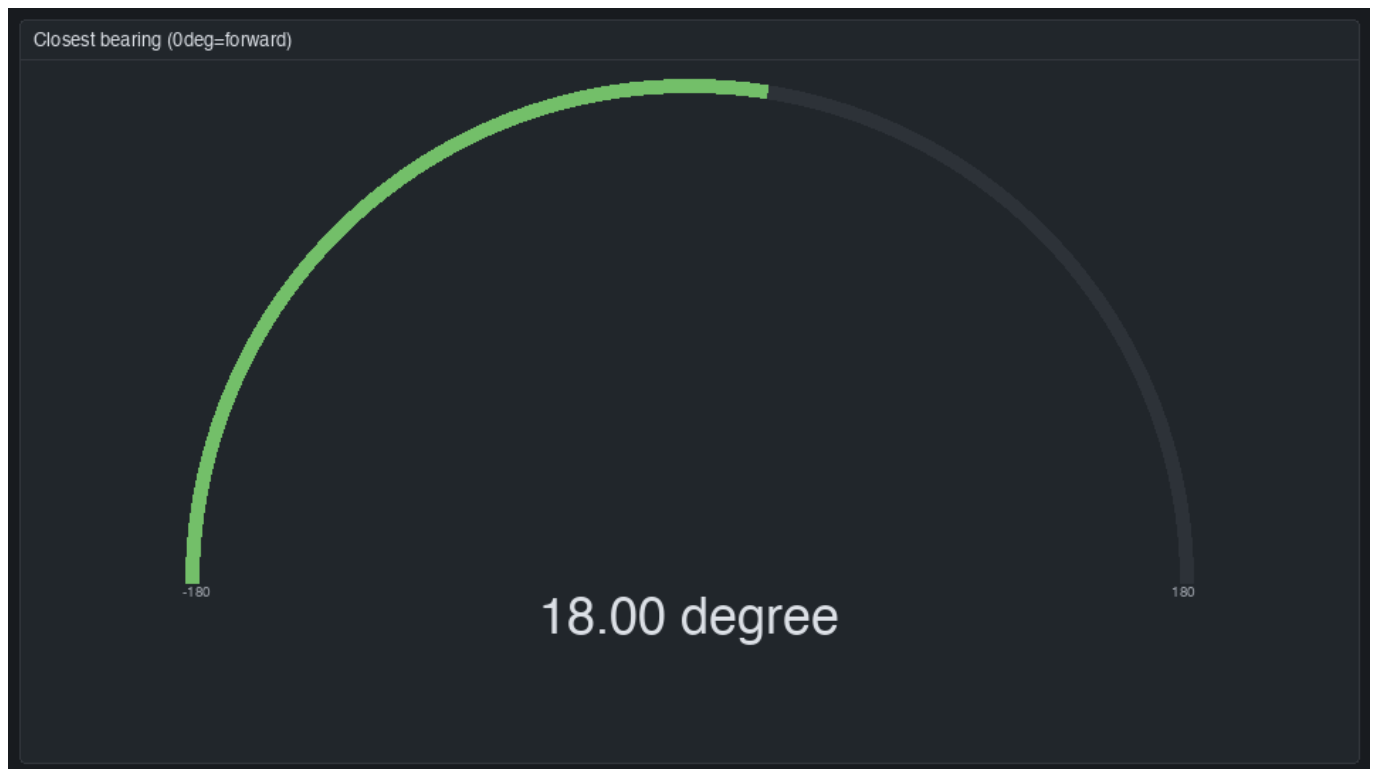


Row 4: LiDAR

Closest obstacle

Row 4: LiDAR



Closest bearing (0deg=forward)*Row 4: LiDAR*

Obstacles < 1 m
Row 4: LiDAR



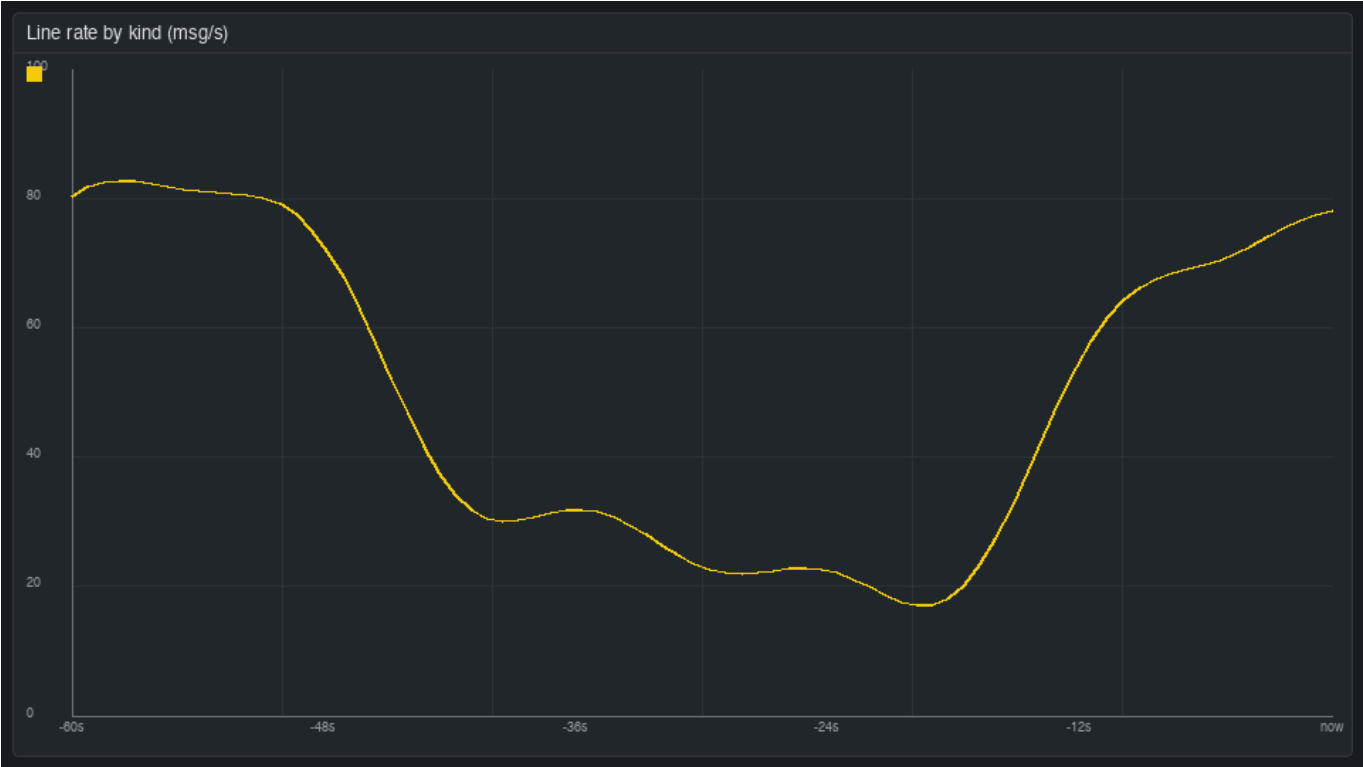
Scan range min / mean / max

Row 4: LiDAR



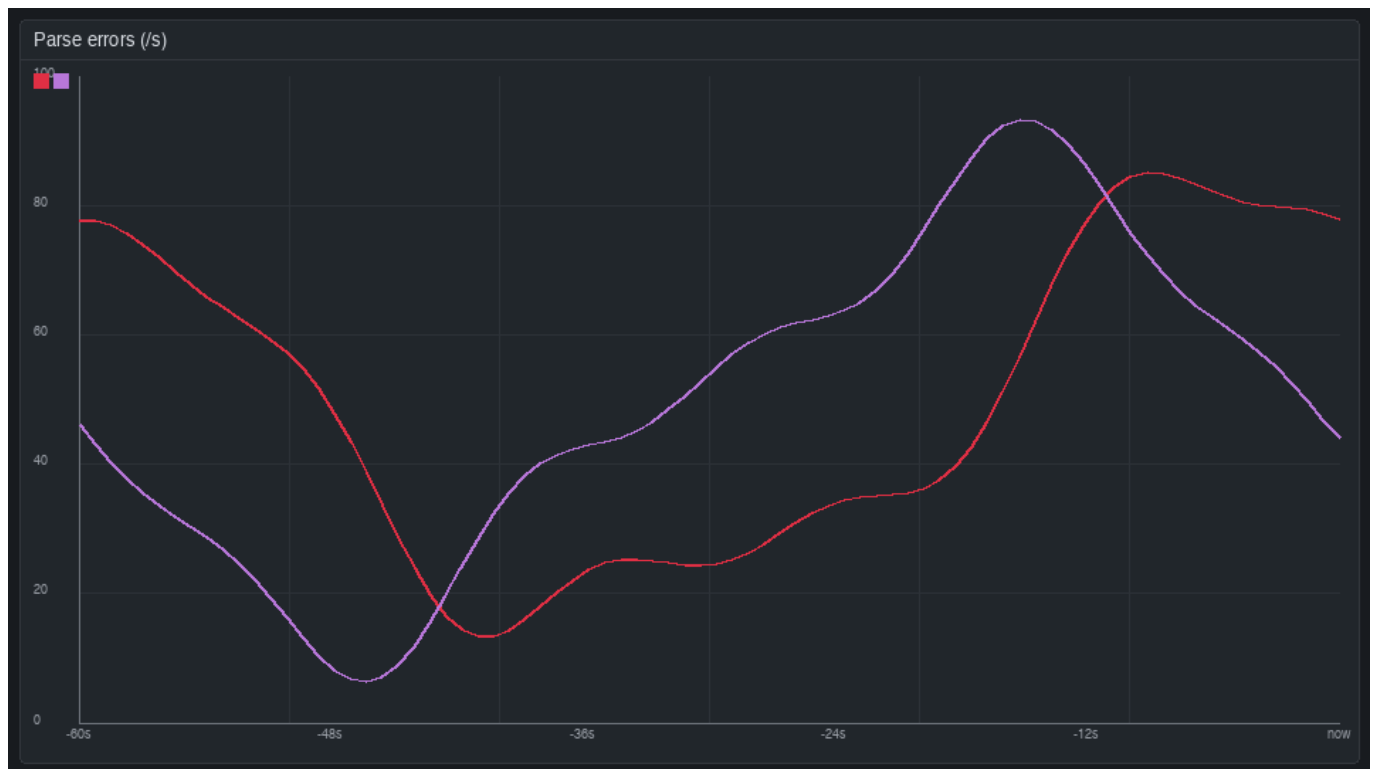
Row 5: Bridge health

Line rate by kind (msg/s)
Row 5: Bridge health

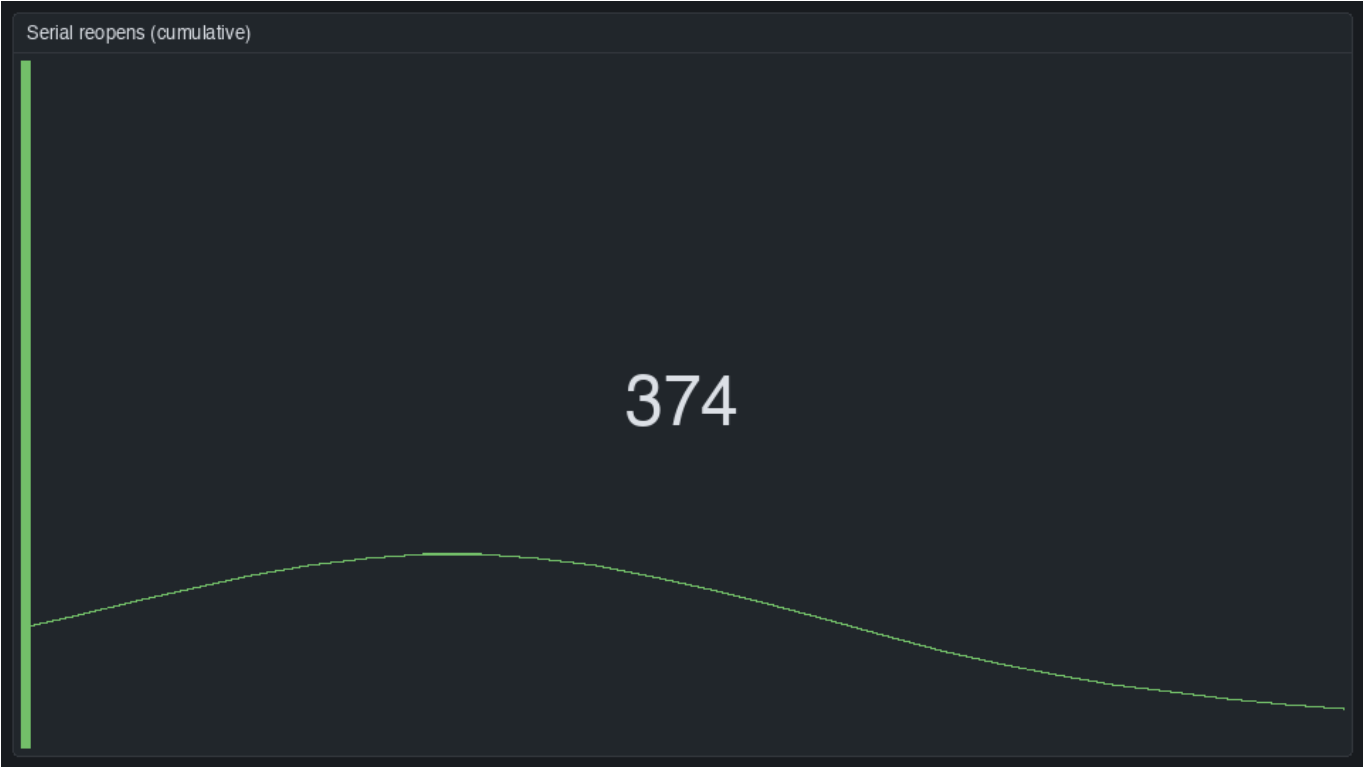


Parse errors (/s)

Row 5: Bridge health

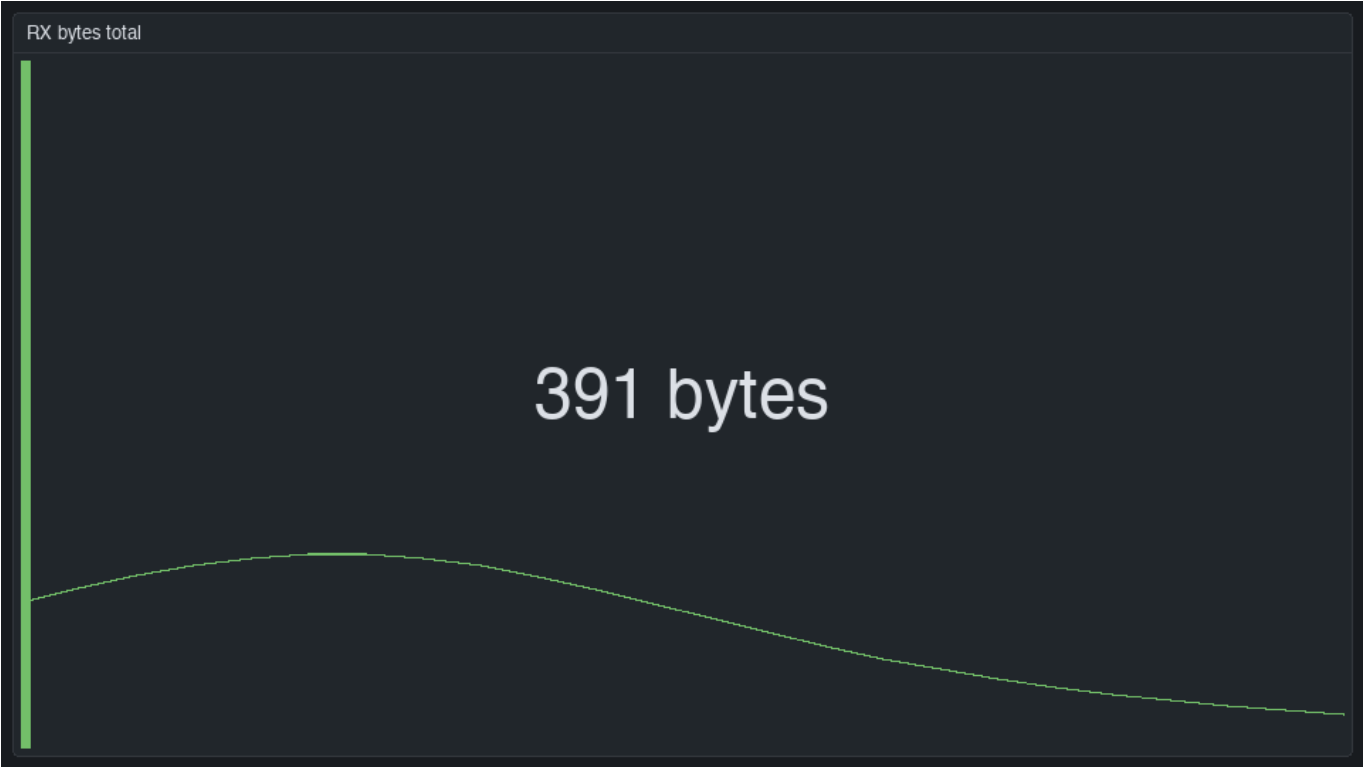


Serial reopens (cumulative)
Row 5: Bridge health



RX bytes total

Row 5: Bridge health



STAR Pi5 Cockpit + Systemd Units

Source Listing

Auto-generated syntax-highlighted listing

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STAR Development Team
Locked Inc.

Generated by `scripts/codedump-pdf.sh`.

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1 (root)

README.md

```
1 # STAR infra -- Pi5-side system files
2
3 Non-ROS infra I manage on the Pi5 itself (outside the ROS workspace). These
4 are mirrored here so they live in version control, but the authoritative
5 copies are the ones deployed at the paths below.
6
7 ## Layout
8
9 ...
10 infra/pi5/
11 +-- opt/
12 |   +-- star_cockpit_api/
13 |   +-- server.py           -> /opt/star_cockpit_api/server.py
14 +-- etc/
15     +-- caddy/
16     |   +-- Caddyfile       -> /etc/caddy/Caddyfile
17     +-- systemd/system/
18         +-- caddy.service   -> /etc/systemd/system/caddy.service
19         +-- star-cockpit-api.service -> /etc/systemd/system/star-cockpit-api.service
20         +-- star-slam-mvp.service -> /etc/systemd/system/star-slam-mvp.service
21     ...
22
23 ## What each piece does
24
25 ### `star_cockpit_api` -- HTTP API backing the Grafana cockpit controls
26
27 A standalone Python rclpy node + HTTP server on `100.64.0.6:9102`.
28
29 Endpoints:
30
31 | Method | Path | Purpose |
32 |-----|-----|-----|
33 | GET    | `/api/state` | `{autonomy, estop, speed, slam}` -- all cached |
34 | POST   | `/api/autonomy/toggle` | flip `/star/autonomy_enable` |
35 | POST   | `/api/estop/toggle` | flip `/star/estop` |
36 | POST   | `/api/speed` | body `{"value":0.75}` -> publish `/star/speed_scale` |
37 | POST   | `/api/cmd_vel` | body `{"linear":0.5,"angular":0}` -> publish `/cmd_vel` |
38 | POST   | `/api/slam/start` | configure+activate `/slam_toolbox` |
39 | POST   | `/api/slam/stop` | deactivate `/slam_toolbox` |
40
41 Subscribes to the latched `/star/state/*` topics so the toggle logic always
42 sees the supervisor's current view. A background thread refreshes the slam
43 lifecycle state every 4s (via `ros2 service call`) so `/api/state` is 0(1).
44
45 ### Systemd units
46
47 - **`caddy.service`** -- Caddy (on Pi5) listens on `100.64.0.6:443`, terminates
48   TLS with its own local CA (`tls internal`), fronts `star_simple_bridge` on
49   `:9101` and `star_cockpit_api` on `:9102`. `After=tailscaled.service` so it
50   waits for the tailscale IP to be assigned. `Restart=always`.
51
52 - **`star-cockpit-api.service`** -- runs `server.py` as the `star` user, with
53   ROS workspace sourced and Cyclone DDS config exported. Restart on failure.
54
55 - **`star-slam-mvp.service`** -- runs `scripts/slam-mvp.sh start` at boot
56   (after tailscale is up + an 8s USB-enumeration delay). Initial state is
57   MANUAL mode, e-stop cleared -- so the robot is safe on boot, operator
58   flips to AUTONOMY via the Grafana cockpit toggle.
59
60 ### `Caddyfile`
61
62 Pi5-side Caddy config. Uses `tls internal` -- no ACME, no Cloudflare, no
63 external CA. Laptop/phone clients install Caddy's local root CA once (see
64 `docs/ARCHITECTURE.md` -> "Client setup to use `star.local`"). No secrets.
65
66 ## Installing on a fresh Pi5
67
68 ```bash
69 # 1. Copy files (from repo root)
70 sudo cp infra/pi5/opt/star_cockpit_api/server.py /opt/star_cockpit_api/
71sudo chmod +x /opt/star_cockpit_api/server.py
72
73sudo cp infra/pi5/etc/caddy/Caddyfile /etc/caddy/Caddyfile
74sudo cp infra/pi5/etc/systemd/system/*.service /etc/systemd/system/
75
```

```

76 # 2. Install Caddy binary with no plugins needed (tls internal only)
77 sudo curl -fsSL -o /usr/local/bin/caddy \
78 "https://caddyserver.com/api/download?os=linux&arch=arm64"
79 sudo chmod +x /usr/local/bin/caddy
80 sudo useradd --system --shell /usr/sbin/nologin --home-dir /var/lib/caddy caddy 2>/dev/null || true
81 sudo mkdir -p /var/log/caddy /var/lib/caddy
82 sudo chown caddy:caddy /var/log/caddy /var/lib/caddy
83
84 # 3. Enable + start
85 sudo systemctl daemon-reload
86 sudo systemctl enable --now caddy star-cockpit-api star-slam-mvp
87 ...
88
89 ## What is deliberately NOT mirrored here
90
91 - **Cluster-side k8s manifests** (traefik, robot-gateway, lichtblick, grafana
92   dashboard JSON). Those live in k3s's etcd; Grafana dashboard is pushed via
93   API. They can be exported separately if needed.
94 - **Any hostname or credentials** -- these system files use only `star.local`
95   and `100.64.0.6` (tailscale IP), nothing that identifies the operator.
96 - **The Caddy-gateway Caddyfile block on pve** that defines
97   `` -- that contains the public hostname and a bcrypt
98   password hash, and is managed on the homelab host, not on the Pi5.

```

2 pi5

pi5/etc/systemd/system/caddy.service

```

1 [Unit]
2 Description=Caddy
3 After=network-online.target tailscaled.service
4 Wants=network-online.target tailscaled.service
5
6 [Service]
7 User=caddy
8 Group=caddy
9 ExecStart=/usr/local/bin/caddy run --config /etc/caddy/Caddyfile
10 ExecReload=/usr/local/bin/caddy reload --config /etc/caddy/Caddyfile --force
11 Restart=always
12 RestartSec=5s
13 TimeoutStopSec=5s
14 LimitNOFILE=1048576
15 AmbientCapabilities=CAP_NET_ADMIN CAP_NET_BIND_SERVICE
16
17 [Install]
18 WantedBy=multi-user.target

```

pi5/etc/systemd/system/star-cockpit-api.service

```

1 [Unit]
2 Description=STAR cockpit HTTP API (Grafana toggle/speed backend)
3 After=network-online.target
4
5 [Service]
6 User=star
7 Group=star
8 ExecStart=/bin/bash -c "source /opt/ros/jazzy/setup.bash && source
9 ↪ /workspaces/STAR/star-ros2/install/setup.bash && export RMW_IMPLEMENTATION=rmw_cyclonedds_cpp && export
10 ↪ CYCLONEDDS_URI=file:///workspaces/STAR/config/cyclonedds.xml && exec python3
11 ↪ /opt/star-cockpit-api/server.py"
12 Restart=always
13 RestartSec=3
14
15 [Install]
16 WantedBy=multi-user.target

```

pi5/etc/systemd/system/star-slam-mvp.service

```

1 [Unit]
2 Description=STAR SLAM MVP stack (supervisor + simple_bridge + LiDAR + SLAM + foxglove)
3 Documentation=file:///workspaces/STAR/docs/ARCHITECTURE.md
4 After=network-online.target tailscaled.service dev-star\x2dmcu.device
5 Wants=network-online.target tailscaled.service
6

```



```

7 [Service]
8 Type=forking
9 User=star
10 Group=star
11 Environment=HOME=/home/star
12 Environment=USER=star
13 Environment=PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin
14 # Give USB devices time to enumerate after boot
15 ExecStartPre=/bin/sleep 8
16 ExecStart=/bin/bash /workspaces/STAR/scripts/slam-mvp.sh start
17 ExecStop=/bin/bash /workspaces/STAR/scripts/slam-mvp.sh stop
18 # The script backgrounds each node and returns; forking type matches
19 RemainAfterExit=yes
20 Restart=no
21
22 [Install]
23 WantedBy=multi-user.target

```

pi5/opt/star_cockpit_api/server.py

```

1  #!/usr/bin/env python3
2  """HTTP API backing the Grafana cockpit controls.
3
4  GET /api/state          -> {"autonomy": bool/null, "estop": bool/null, "speed": float/null}
5  POST /api/autonomy/toggle -> flips /star/autonomy_enable, returns new state
6  POST /api/estop/toggle  -> flips /star/estop, returns new state
7  POST /api/speed body: {"value": 0.75} -> publishes /star/speed_scale, returns applied
8
9  Binds to 127.0.0.1:9102; Caddy fronts it as https://star.local/api/*
10 State is learned from the supervisor's latched /star/state/* topics.
11 """
12 import json
13 import subprocess
14 import re
15 import threading
16 from http.server import BaseHTTPRequestHandler, ThreadingHTTPServer
17 import rclpy
18 from rclpy.node import Node
19 from rclpy.qos import QoSProfile, DurabilityPolicy, ReliabilityPolicy, HistoryPolicy
20 from std_msgs.msg import Bool, Float32
21 from geometry_msgs.msg import Twist
22
23 PORT = 9102
24 LATCHED = QoSProfile(
25     history=HistoryPolicy.KEEP_LAST, depth=1,
26     reliability=ReliabilityPolicy.RELIABLE,
27     durability=DurabilityPolicy.TRANSIENT_LOCAL,
28 )
29
30
31 class CockpitNode(Node):
32     def __init__(self):
33         super().__init__("star_cockpit_api")
34         self._lock = threading.Lock()
35         self._autonomy = None
36         self._estop = None
37         self._last_speed = None
38         self._slam_state_cached = None
39         self._autonomy_pub = self.create_publisher(Bool, "/star/autonomy_enable", 10)
40         self._estop_pub = self.create_publisher(Bool, "/star/estop", 10)
41         self._speed_pub = self.create_publisher(Float32, "/star/speed_scale", 10)
42         self._cmd_vel_pub = self.create_publisher(Twist, "/cmd_vel", 10)
43         self.create_subscription(Bool, "/star/state/autonomy_enable", self._on_auto, LATCHED)
44         self.create_subscription(Bool, "/star/state/estop", self._on_estop, LATCHED)
45         self.get_logger().info(f"cockpit HTTP on 127.0.0.1:{PORT}")
46
47     def _on_auto(self, m):
48         with self._lock: self._autonomy = bool(m.data)
49
50     def _on_estop(self, m):
51         with self._lock: self._estop = bool(m.data)
52
53     def get_state(self):
54         with self._lock:
55             return {"autonomy": self._autonomy, "estop": self._estop, "speed": self._last_speed, "slam":
56                 ↪ self.slam_state()}
57
58     def toggle_autonomy(self):

```

```

58         with self._lock:
59             new = not bool(self._autonomy)
60             self._autonomy_pub.publish(Bool(data=new))
61             return new
62
63     def toggle_estop(self):
64         with self._lock:
65             new = not bool(self._estop)
66             self._estop_pub.publish(Bool(data=new))
67             return new
68
69     def slam_lifecycle(self, transition_id):
70         try:
71             r = subprocess.run(
72                 ["ros2", "service", "call", "/slam_toolbox/change_state",
73                  "lifecycle_msgs/srv/ChangeState",
74                  "{transition: {id: %d}}" % transition_id],
75                 capture_output=True, text=True, timeout=15)
76             ok = "success=True" in r.stdout
77             self._refresh_slam_state()
78             return ok
79         except Exception:
80             return False
81
82     def slam_state(self):
83         return self._slam_state_cached
84
85     def _refresh_slam_state(self):
86         try:
87             r = subprocess.run(
88                 ["ros2", "service", "call", "/slam_toolbox/get_state",
89                  "lifecycle_msgs/srv/GetState"],
90                 capture_output=True, text=True, timeout=5)
91             m = re.search(r"label=\'([a-z_]+)\'", r.stdout)
92             new = m.group(1) if m else None
93             with self._lock:
94                 self._slam_state_cached = new
95         except Exception:
96             pass
97
98     def set_cmd_vel(self, linear, angular):
99         m = Twist(); m.linear.x = float(linear); m.angular.z = float(angular)
100         self._cmd_vel_pub.publish(m)
101         return {"linear": m.linear.x, "angular": m.angular.z}
102
103     def set_speed(self, v):
104         v = max(0.0, min(1.0, float(v)))
105         with self._lock:
106             self._last_speed = v
107         self._speed_pub.publish(Float32(data=v))
108         return v
109
110
111     NODE = None # set in main
112
113
114     class H(BaseHTTPRequestHandler):
115         def log_message(self, *a, **k): pass
116
117         def _json(self, code, obj):
118             body = json.dumps(obj).encode()
119             self.send_response(code)
120             self.send_header("Content-Type", "application/json")
121             self.send_header("Content-Length", str(len(body)))
122             self.send_header("Cache-Control", "no-store")
123             self.send_header("Access-Control-Allow-Origin", "*")
124             self.send_header("Access-Control-Allow-Methods", "GET, POST, OPTIONS")
125             self.send_header("Access-Control-Allow-Headers", "Content-Type")
126             self.end_headers()
127             self.wfile.write(body)
128
129         def do_OPTIONS(self):
130             self._json(204, {})
131
132         def do_GET(self):
133             if self.path == "/api/state":
134                 return self._json(200, NODE.get_state())
135             self._json(404, {"error": "not found"})
136

```

```

137 def do_POST(self):
138     if self.path == "/api/autonomy/toggle":
139         return self._json(200, {"autonomy": NODE.toggle_autonomy()})
140     if self.path == "/api/estop/toggle":
141         return self._json(200, {"estop": NODE.toggle_estop()})
142     if self.path == "/api/slam/start":
143         ok = NODE.slam_lifecycle(3)
144         if not ok:
145             NODE.slam_lifecycle(1)
146             ok = NODE.slam_lifecycle(3)
147         return self._json(200 if ok else 500, {"slam": NODE.slam_state(), "ok": ok})
148     if self.path == "/api/slam/stop":
149         ok = NODE.slam_lifecycle(4)
150         return self._json(200 if ok else 500, {"slam": NODE.slam_state(), "ok": ok})
151     if self.path == "/api/cmd_vel":
152         n = int(self.headers.get("Content-Length", 0) or 0)
153         try:
154             body = json.loads(self.rfile.read(n).decode() or "{}")
155             lin = float(body.get("linear", 0))
156             ang = float(body.get("angular", 0))
157         except Exception as e:
158             return self._json(400, {"error": f"bad body: {e}"})
159         return self._json(200, NODE.set_cmd_vel(lin, ang))
160     if self.path == "/api/speed":
161         n = int(self.headers.get("Content-Length", 0) or 0)
162         try:
163             v = float(json.loads(self.rfile.read(n).decode() or "{}").get("value"))
164         except Exception as e:
165             return self._json(400, {"error": f"bad body: {e}"})
166         return self._json(200, {"speed": NODE.set_speed(v)})
167     self._json(404, {"error": "not found"})
168
169
170 def _slam_refresher_loop():
171     import time
172     while True:
173         try:
174             if NODE is not None:
175                 NODE._refresh_slam_state()
176         except Exception:
177             pass
178         time.sleep(4)
179
180 def main():
181     global NODE
182     rclpy.init()
183     NODE = CockpitNode()
184     srv = ThreadingHTTPServer(("100.64.0.6", PORT), H)
185     threading.Thread(target=srv.serve_forever, name="cockpit-http", daemon=True).start()
186     threading.Thread(target=_slam_refresher_loop, name="slam-refresh", daemon=True).start()
187     try:
188         rclpy.spin(NODE)
189     finally:
190         NODE.destroy_node()
191         rclpy.shutdown()
192
193
194 if __name__ == "__main__":
195     main()

```